Hyperspectral Anomaly Detection via MERA Decomposition and Enhanced Total Variation Regularization

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***Abstract*— In recent years, tensor representation (TR)-based hyperspectral anomaly detection (HAD) approaches have attracted more and more attention. However, two urgent issues still need to be addressed: 1) existing tensor decomposition approaches for HAD cannot make full use of the spectral–spatial correlation of background components in hyperspectral images (HSIs) and 2) most approaches based on TR overlook the piecewise-smooth of background components that exist simultaneously in the spectral and spatial domains. To this end, with the aid of an advanced multiscale entanglement renormalization ansatz (MERA) tensor network, this article proposes an algorithm based on MERA decomposition and enhanced total variation regularization (MERAETV) for HAD. Specifically, MERA decomposes the background tensor by contracting a top-level factor with the remaining semiorthogonal and orthogonal factors. Due to the intricate interplay between semiorthogonal (low-rank) and orthogonal factors, low-rank MERA approximation exhibits a robust representational capacity that effectively captures the spectral–spatial correlation of the background component. Meanwhile, an enhanced total variation (ETV) regularization is devised to capture the inherent piecewise-smooth of the background component in both spectral and spatial domains. Furthermore, our algorithm incorporates group sparsity constraint and Gaussian noise term to enhance the discrimination between anomalies and background. Finally, a highly efficient update scheme based on the alternating direction method of multipliers (ADMM) is designed. A large number of experiments on one synthetic and seven real HSIs demonstrate the superiority of our proposed approach.**

***Index Terms*— Anomaly detection, enhanced total variation (ETV), hyperspectral images (HSIs), low-rank multiscale entanglement renormalization ansatz (MERA) approximation, MERA decomposition.**

NOMENCLATURE

b	Scalar.
$\boldsymbol{B}$	Matrix.
$\boldsymbol{B}_{ij}$ or b_{ij}	(i, j)th entry.
$\boldsymbol{B}^{-1}$	Inverse of $\boldsymbol{B}$.
$\mathrm{Tr}(\boldsymbol{B})$	Trace of $\boldsymbol{B}$.
$\|\boldsymbol{B}\|_F$	$(\sum_{ij} \|b_{ij}\|^2)^{1/2}$.
$\boldsymbol{b}$	Vector.
$\boldsymbol{\mathcal{B}}$	Tensor.
$\boldsymbol{\mathcal{B}}_{ijk}$ or b_{ijk}	(i, j, k)th entry.
$\boldsymbol{\mathcal{B}}^*$	Conjugate transpose of $\boldsymbol{\mathcal{B}}$.
$\boldsymbol{B}_{(i)}$	Mode-i matricization of $\boldsymbol{\mathcal{B}}$.
$\|\boldsymbol{\mathcal{B}}\|_F$	$(\sum_{ijk} \|b_{ijk}\|^2)^{1/2}$.

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) contain abundant spatial details and numerous nearly continuous spectral bands. These abundant spatial information and precise spectral features provide a scientific theoretical basis for the diagnosis and identification of different ground objects, rendering HSIs extensively employed in various fields, including change detection [1], spectral unmixing [2], and anomaly detection [3]. Recently, there has been a growing interest in hyperspectral anomaly detection (HAD) due to its significant applications in both civilian and military fields. The purpose of HAD is to locate abnormal targets that exhibit notable spectral deviations from the surrounding background without prior information on the spectrum of ground objects. Based on the adoption of various theoretical knowledge, current HAD methods can be broadly categorized into approaches based on statistical theory [4], [5], [6], [7], approaches based on deep learning [8], [9], [10], and approaches based on representation [11], [12], [13], [14], [15], [16]. In this article, we focus on representation-based approaches.

As a natural representation of high-dimensional data, tensors can effectively express features in high-dimensional space and maintain the integrity of the feature structure. For HAD problems, tensor representation (TR) has natural advantages. There are generally two methods for low-rank characterization of high-dimensional data: one is minimizing tensor rank and the other is low-rank tensor decomposition [17]. The definition of tensor rank is different from the definition of matrix rank. The difference is that the definition of tensor rank is not unique. Usually, the definition of tensor rank is determined by the corresponding tensor decomposition form [18]. According to the different forms of tensor decomposition, existing techniques can be broadly categorized into four groups: approaches based on CANDECOMP/PARAFAC (CP) decomposition [19],

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[20], approaches based on Tucker decomposition [21], [22], approaches based on tensor singular value decomposition (t-SVD) [23], [24], [25], [26], [27], and approaches based on tensor ring decomposition (TRD) [28]. CP rank is defined as the minimum number of rank-one tensors required to express a tensor using CP decomposition, which is usually NP-hard to compute [29]. Tucker rank is defined as a vector whose elements are the ranks of all expanded matrices along modulo-n ($n = 1, 2, 3$) when expressing a tensor using Tucker decomposition. Since matrix operation is involved, the inherent multidimensional structure information of tensor data may be destroyed [30]. To alleviate it, the tube rank of a tensor based on t-SVD is defined, where the tubal rank is the number of nonzero tube ranks in the middle diagonal tensor in t-SVD. As a convex relaxation of the tensor tube rank, the tensor nuclear norm (TNN) is used to estimate the low-rank of the background tensor [23], [25], [31]. Nevertheless, various nuclear norms developed based on t-SVD still rely on each independent tensor slice, resulting in the inability to fully utilize the advantages of the spectral–spatial dimensional structural information of the entire tensor [32]. In order to address this issue, Feng et al. [28] came up with TRD to further mine the low-rank characteristics of the background tensor existing in each dimension in HSI by decomposing a third-order background tensor into a multilinear tensor product of three factor tensors. However, TRD only establishes a connection between two adjacent factor tensors, resulting in limited correlation representation of high-order tensors. Furthermore, the estimation of TRD rank is still an open issue [33].

Apart from investigating the low-rank characteristics of the background, several studies have further delved into exploring the piecewise-smooth of the background to comprehensively improve detection results [20], [24], [25], [34], [35]. For instance, Li et al. [24] designed a PTA algorithm based on prior, leveraging the low-rank property of the background in the spectral domain and the spatial smoothness of the background to enhance detection accuracy. Nevertheless, PTA does not consider the low-rank nature of the background in the spatial domain and the piecewise-smooth in the spectral domain, which weakens the detection performance. The low average rank total variation anomaly detection (LARTVAD) algorithm provided by Sun et al. [25] utilizes TNN and total variation (TV) regularization constraints to explore the low-rank nature and the piecewise-smooth of the background. However, LARTVAD fails to consider the smoothness of the background in the spectral domain. In general, most of the abovementioned algorithms focus on the piecewise-smooth of the background existing in the spatial dimension.

Based on the above analysis, although some TR-based detection methods exhibit relatively satisfactory detection results, there are still two key issues that need to be resolved.

1) When facing high-dimensional data, the current tensor decomposition techniques used for HAD still have limitations and cannot fully mine the spectral–spatial correlation of the relatively complex background. Therefore, there are always unavoidable errors in the accurate estimation of low-rank background.
2) Most of the existing TR methods only explore the piecewise-smooth of the background component in the spatial dimension but ignore the piecewise-smooth of the background component in the spectral dimension. When the target does not belong to low-frequency information, insufficient mining of background components may cause the loss of texture and detailed features of small targets.

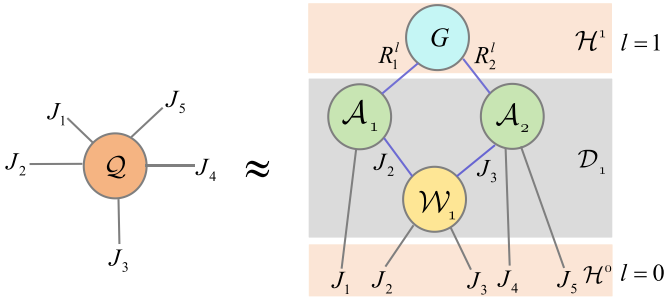


Fig. 1. Schematic illustration of the MERA decomposition for a fifth-order tensor.

In order to simultaneously address the aforementioned challenges, a model based on multiscale entanglement renormalization ansatz (MERA) decomposition and enhanced TV (ETV) regularization is developed, referred to as MERAETV. MERAETV mainly consists of MERA decomposition and ETV regularization, and its basic motivation can be summarized as follows.

1) Inspired by TRD [28], a low-rank stable tensor network MERA decomposition is introduced in this article [33], [36], [37]. MERA is a class of tree-like tensor networks that can be represented in a hierarchical manner while maintaining computational advantages. Fig. 1 shows a schematic of the MERA decomposition of a fifth-order tensor. Specifically, MERA decomposes a fifth-order tensor into contractions of a top-level core factor and the remaining orthogonal ($\boldsymbol{\mathcal{W}}_1$) factor and two semiorthogonal ($\boldsymbol{\mathcal{A}}_1, \boldsymbol{\mathcal{A}}_2$) factors. Due to the multiple interactions between orthogonal ($\boldsymbol{\mathcal{W}}_1$) factor and semiorthogonal ($\boldsymbol{\mathcal{A}}_1, \boldsymbol{\mathcal{A}}_2$) (low-rank) factors, low-rank MERA can fully mine the correlation between high-order tensor data dimensions, thereby providing a more compact and expressive low-rank approximate representation for high-order tensor. In fact, Long et al. [32] have demonstrated that for RGB images, under the same compression ratio, the low-rank MERA approximation exhibits better low-rank recovery performance than other approximation approaches (t-SVD and Tucker). As an advanced tensor decomposition method, MERA has demonstrated a unique advantage in mining the correlation of high-order tensor data and has been successfully applied in fields such as quantum mechanics [38] and multiview clustering [32]. Based on such success, we are motivated to utilize the MERA framework in HAD tasks. In view of the cubic characteristics of HSI data, an HAD model based on low-rank MERA approximation is designed. MERA decomposition is used to mine the deep-level features of the background component in the HSI cube to adapt to the variability of the scene and fully capture the spectral–spatial structure information, thereby achieving high-precision detection.

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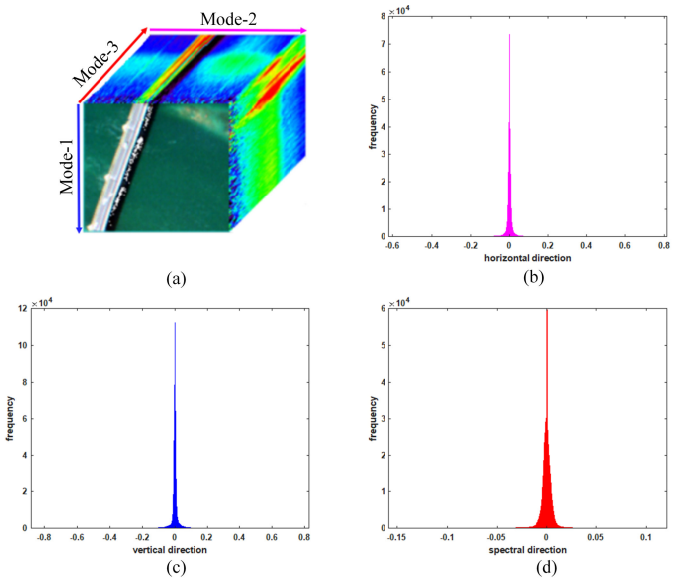


Fig. 2. Illustration of the TV properties of HSI tensor data in both spectral and spatial dimensions. (a) HSI tensor data. (b) Differential statistics in the horizontal direction. (c) Differential statistics in the vertical direction. (d) Differential statistics in the spectral direction.

2) TV regularization is used in HAD to characterize the piecewise-smooth of the background. For HSI tensor data, the piecewise-smooth of both spectral and spatial domains has been verified in HSI restoration [39]. However, the HAD method based on TR mostly only considers the segmented smoothness of the background in the spatial domain while neglecting the segmented smoothness of the background in the spectral domain, such as [20], [24], [25], [34], and [35]. To illustrate this point more clearly, Fig. 2 shows the TV characteristics of an HSI tensor data that exist in both spectral and spatial dimensions. It can be easily seen that HSI also has a local smooth structure along its spectral direction, that is, the differences between adjacent bands in the spectral domain are mostly close to zero. It is obvious that the traditional TV regularization ignores the piecewise-smooth of the spectral domain, which encourages us to design a new ETV regularizer based on the tensor model to fully exploit the piecewise-smooth of the background that exists in both the spectral and spatial domains simultaneously.

In summary, this article provides the following main contributions.

1) We develop a novel TR-based algorithm, referred to as MERAETV. MERAETV simultaneously considers the global correlation and local piecewise-smooth of the background in the spectral and spatial domains, as well as the sparseness of the anomalies and the robustness to noise.
2) Thanks to the robust representation capability of MERA decomposition, we introduce the low-rank MERA approximation into the MERAETV algorithm to effectively exploit the spectral–spatial correlation of the background in HSI.
3) An ETV regularizer is designed and incorporated into the MERA decomposition framework to effectively exploit the local piecewise-smooth and global correlation of the background tensor in the spectral and spatial domains, thereby making the estimated background closer to the actual background. In addition, group sparsity constraint and Gaussian noise term are also incorporated into the MERAETV algorithm to enhance the discrimination of anomalies and background.
4) An optimization algorithm based on the alternating direction method of multipliers (ADMM) is derived to efficiently solve the MERAETV model. Extensive experiments demonstrate that the anomaly detection performance of our proposed approach outperforms the existing state-of-the-art approaches.

The remainder of this article is organized as follows. Section II reviews some related work. In Section III, some frequently employed symbols and preliminaries are presented. The proposed MERAETV model and optimization solution process are introduced in detail in Section IV. Section V shows extensive experimental results to verify the superiority of the proposed model. Finally, the work is summarized in Section VI.

II. Related Work

In this section, some representation-based HAD methods will be introduced. Representation-based approaches can be further divided into matrix representation (MR)-based approaches [11], [12], [13], [14], [40], [41], [42], [43] and TR-based approaches [19], [20], [21], [22], [23], [35], [44], [45].

A. MR-Based Approaches

Approaches based on MR mainly include collaborative representation (CR)-based approaches [11], [40], [41], sparse representation (SR)-based approaches [13], [42], and low-rank representation (LRR)-based approaches [12], [14], [43]. The CRD anomaly detector developed by Li and Du [11] is one of the pioneering works based on CR methods. This approach suggests that pixels in the background can be represented linearly by pixels within double windows arranged concentrically, whereas abnormal pixels cannot. However, when the abnormal target is large or the background is complex, the background dictionary constructed by sliding concentric dual windows is easily interfered with by abnormal targets, resulting in unsatisfactory detection results. To address this issue, various CR-based variant algorithms have been designed [40], [41], [46], [47]. Similarly, SR-based methods assume that the pixel to be measured can be linearly represented by a few pixels in an overcomplete dictionary. The background joint SR (BJSR) is a typical method based on SR, which leverages orthogonal subspace to acquire background basis and subsequently detects anomalous objects [13].

Different from the first two types of approaches, the LRR-based approaches assume that background and anomalies are low-rank and sparse, respectively [12]. Depending on whether it is necessary to construct a background dictionary, LRR-based approaches can be subdivided into methods based

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on low-rank and SR (LRaSR) [48], [49], [50], [51], [52], [53], [54], [55], [56], [57] and methods based on low-rank and sparse matrix decomposition (LRaSMD) [14], [43], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67]. Among them, the former requires the construction of a background dictionary. The core idea of the LRaSR-based methods is to model the background as the product of a background dictionary and a coefficient matrix, and characterize the background by imposing various regularization constraints on the coefficient matrix. For example, in [50], the background is characterized by simultaneously imposing low-rank and sparse dual constraints on the coefficient matrix. Cheng and Wang [51] introduced a graph and TV-regularized LRR (GTVLRR) algorithm by imposing graph regularization and TV constraints on the coefficient matrix. Feng et al. [52] developed a local space and TV constraint (LSC-TV) algorithm by applying local space constraint and TV regularization to the coefficient matrix. Instead of the TV constraint directly imposed on the coefficient matrix, Zhao et al. [53] designed an ETV used on the row vectors of the coefficient matrix to enhance the spatial structure of HSI in the representation process. In order to improve detection performance, Wang et al. [54] integrated the reweighted nuclear norm and TV into the LRaSR model and constructed a sparse dictionary that does not require clustering technology. Considering the impact of noise on detection performance, Wang et al. [55] came up with a dual subspace representation learning (DSR-ADNR) algorithm for joint denoising and detection, which can simultaneously complete image denoising and detect anomalous targets. Furthermore, in order to obtain a good background representation, a hierarchical alternating optimization scheme is designed in [56], as well as improving the antinoise performance by modeling mixed noise. In [57], starting from the unmixing perspective, accurate abundance and background endmembers are learned to represent the background for anomaly detection. In general, the detection performance of the abovementioned algorithms depends largely on the quality of the constructed background dictionary.

Compared with the LRaSR-based approaches, the core idea of the LRaSMD-based methods is to decompose HSI data directly into three components: background matrix, anomaly matrix, and noise matrix; the anomaly detection is realized by applying various prior constraints to the three components, respectively. An early approach is robust principal component analysis (RPCA), which decomposes HSI data into a sparse anomaly component and a low-rank background component [14]. In order to reduce the influence of noise, a GoDec-based anomaly detection method called LSMAD is proposed [43]. LSMAD introduces an additional noise term in the decomposition process, where the noise of the image is estimated through GoDec, and then, GRX is applied to the sparse matrix to detect abnormal targets. However, the GoDec-based method is still insufficient in finding the rank of low-rank backgrounds and the sparse cardinality of sparse anomalies. In order to solve this problem, in [58], a GoDec method based on orthogonal subspace projection was developed and achieved satisfactory detection performance. Recently, similar methods have been fully extended in [59] and [60]. In [61], the sparse component was modeled as a mixture of Gaussian, and an LSDM-MoG method was designed to distinguish anomalies and noise. On this basis, Feng et al. [62] comprehensively utilized the information in low-rank component and sparse components by introducing a density peak-guided CR method. In [63], the fractional Fourier transform was introduced into the LRaSMD model, and a fractional enhanced background purification anomaly detection (FEBPAD) algorithm was provided. Chen et al. [64] proposed a new method called component decomposition analysis (CDA), which decomposes HSI data into three parts, principal component, independent component, and noise component, and achieved competitive performance. In [65], a data spheroidization technique was used to remove the background instead of using a low-rank space to represent the background. Recently, in [66], an HAD method utilizing band sampling in the effective anomaly space was proposed, which focused on solving the problem of interband correlation reduction. In [67], a new method called FBGVLRS-AD is presented, which decomposes full-band data into multiple small-scale band groups and performs anomaly detection between each band group. Then, taking advantage of the consistency and complementarity of anomaly detection results between different band groups, an iterative feedback framework is designed to improve detection performance. However, the aforementioned HAD algorithms all involve the conversion of a 3-D HSI cube into a 2-D matrix for processing, resulting in an inevitable loss of crucial structural information inherent in the hyperspectral cube.

B. TR-Based Approaches

In [19], an existing detection model is used to remove potential abnormal pixels, and the low-rank background is reconstructed based on CP decomposition and then detects abnormal targets through the residual between the original HSI and the reconstructed background. Shang et al. [20] approximated the background tensor as the modulo-3 product of the abundance tensor and the endmember matrix and used CP decomposition and TV smoothing constraints to characterize the low-rank background. In order to better utilize the spatial information of HSIs, Feng et al. [21] designed an HAD approach based on TV-regularized low-rank Tucker decomposition and CR. This approach obtains the final detection map by fusing the initial detection results of the low-rank part and the mixed information part. In [22], it was found that the core tensor generated by Tucker decomposition of the gradient tensor exhibits sparsity. Therefore, a regularization term that can simultaneously characterize the low-rank and local smoothness properties of the background is designed to improve detection performance. Based on t-SVD, Xu et al. [23] simultaneously completed compression and anomaly detection of HSIs. Wang et al. [44] introduced the concept of tensor LRR into HAD for the first time, achieving the separation of low-rank background and sparse anomalies. On this basis, in order to obtain an accurate low-rank background estimate, various improved algorithms have been produced [35], [45]. In addition, Qin et al. [68] replaced the convex surrogate TNN with a nonconvex function and developed a generalized nonconvex low-rank tensor approximation model. In [28], TRD and TV regularization constraints are introduced to explore the low-rank and piecewise-smooth of the background. However, the various

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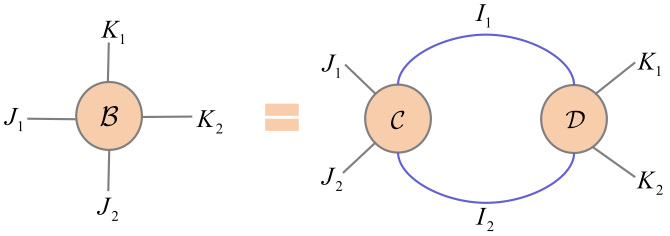


Fig. 3. Schematic of tensor contraction.

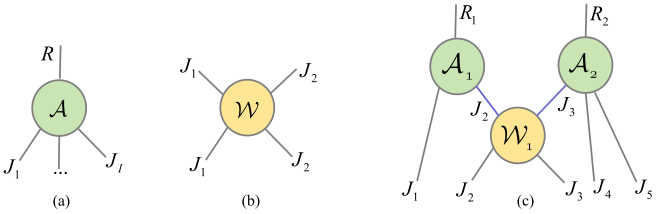


Fig. 4. Schematic of (a) isometry, (b) disentangler, and (c) layer, respectively.

tensor decomposition methods involved in the abovementioned algorithms still have limitations and cannot fully exploit the spectral–spatial correlation in complex backgrounds.

Different from various low-rank tensor decomposition methods involved in the above-mentioned algorithms, this article proposes a low-rank stable tensor network MERA decomposition. It is worth noting that MERA decomposes a high-order background tensor into contractions of a top-level core factor and the remaining orthogonal and semiorthogonal factors. Thanks to the multiple interactions between orthogonal and semiorthogonal (low-rank) factors, low-rank MERA can fully mine the correlation between high-order background tensor data dimensions, thereby obtaining an accurate low-rank background estimate.

Note: The tensor network approach presented in this article and the neural network in deep learning are two types of networks that are derived from different observations and have been successful in different applications. Interestingly, they are intrinsically connected through multilinear structures, rather than being orthogonal to each other [69], [70].

III. Symbols and Preliminaries

Some frequently employed symbols in this article are shown in the Nomenclature. For any given two tensors $\mathcal{C} \in \mathbb{R}^{J_1\times J_2\times I_1\times I_2}$ and $\mathcal{D} \in \mathbb{R}^{I_1\times I_2\times K_1\times K_2}$, their tensor contraction is represented by $\mathcal{B} \in \mathbb{R}^{J_1\times J_2\times K_1\times K_2}$, that is, $\mathcal{B} = \mathcal{C} \times_{\{I_1,I_2\}} \mathcal{D}$, where $b_{j_1,j_2,k_1,k_2} = \sum_{i_1,i_2} c_{j_1,j_2,i_1,i_2} d_{i_1,i_2,k_1,k_2}$, as shown in Fig. 3. Subsequently, we will introduce the two basic components of MERA decomposition: isometry and disentangler.

Definition 1 (Isometry [32], [37]): An isometry $\mathcal{A} \in \mathbb{R}^{J_1\times\cdots\times J_I\times R}$ can be defined as a $(I+1)$th-order tensor with semiorthogonal unfolding matrix $\boldsymbol{A} \in \mathbb{R}^{J_1,\ldots,J_I\times R}$, i.e., $\boldsymbol{A}^*\boldsymbol{A} = \boldsymbol{I}_R$, where I denotes the number of bottom legs and R represents the outgoing dimension of an isometry, as illustrated in Fig. 4(a).

Definition 2 (Disentangler [32], [37]): A disentangler $\mathcal{W} \in \mathbb{R}^{J_1\times J_2\times J_1\times J_2}$ is defined as fourth-order tensor with a square orthogonal unfolding matrix $\boldsymbol{W} \in \mathbb{R}^{J_1J_2\times J_1J_2}$, denoted as $\boldsymbol{W}^*\boldsymbol{W} = \boldsymbol{I}_{J_1J_2}$, as depicted in Fig. 4(b).

Definition 3 (Layer [32], [37]): A layer $\mathcal{D}$ is defined as an outcome of tensor contraction between disentanglers and isometries within the same layer. For instance, as illustrated in Fig. 4(c), a seventh-order layer $\mathcal{D}$ can be constructed by employing two isometries $\mathcal{A}_1 \in \mathbb{R}^{J_1\times J_2\times R_1}$ and $\mathcal{A}_2 \in \mathbb{R}^{J_3\times J_4\times J_5\times R_2}$ and one disentangler $\mathcal{W}_1 \in \mathbb{R}^{J_2\times J_3\times J_2\times J_3}$ as follows:

$$\mathcal{D} = \mathcal{W}_1 \times_{\{J_2\}} \mathcal{A}_1 \times_{\{J_3\}} \mathcal{A}_2 \in \mathbb{R}^{J_1\times J_2\times J_3\times J_4\times J_5\times R_1\times R_2}. \quad (1)$$

Definition 4 (MERA Decomposition [32], [37]): The MERA representation of an Nth-order tensor $\mathcal{Q} \in \mathbb{R}^{J_1\times J_2\times\cdots\times J_N}$ can be constructed through the composition of different layers and a top-level core factor, which can be recursively achieved through the following procedure:

$$\mathcal{H}^{l-1} = \mathcal{H}^l \times_{\{R^l_1,\ldots,R^l_{G_l}\}} \mathcal{D}_l \quad (2)$$

where $R^l_{g_l}, g_l = 1,\ldots,G_l, l = 1,\ldots,L$, represents the MERA ranks. G_l denotes the number of isometries in the lth layer, and L represents the total number of layers. $\mathcal{H}^0 = \mathcal{Q}$, and $\mathcal{H}^L = \boldsymbol{G}$, where $\boldsymbol{G}$ is the top-level core factor. Through a nested manner, the MERA decomposition is constructed.

For the convenience of memory, we use $\mathcal{Q} = f(\boldsymbol{G}, \mathcal{A}, \mathcal{W})$ to denote the MERA decomposition, where $\mathcal{A} = (\mathcal{A}^l_1,\ldots,\mathcal{A}^l_{G_l})$, $\mathcal{W} = (\mathcal{W}^l_1,\ldots,\mathcal{W}^l_{G_l-1})$, and $\mathcal{W}^l_{g_l}$ and $\mathcal{A}^l_{g_l}$ are the g_lth disentangler and isometry, respectively, in the lth layer, where $l = 1,\ldots,L$. Fig. 1 presents a schematic of the MERA decomposition of a fifth-order tensor.

IV. Proposed Method

The flowchart illustrating the MERAETV model proposed in this article is depicted in Fig. 5. First, the original HSI tensor data $\mathcal{X}$ are decomposed into three parts: background component $\mathcal{Z}$, anomaly component $\mathcal{E}$, and noise component $\mathcal{N}$. Then, the background component $\mathcal{Z}$ is redistributed through MERA decomposition to improve the separability of anomalies and background. At the same time, the low-rank MERA approximation is employed to fully mine the spectral–spatial correlation of the background component. In order to take full advantage of the piecewise-smooth of the background component $\mathcal{Z}$ in the spectral and spatial domains, an ETV regularization based on the tensor model is designed. Next, an $\ell_{2,1,1}$-norm is applied to anomaly component $\mathcal{E}$ to promote the group sparsity of anomalous targets. In addition, the noise term $\mathcal{N}$ is modeled as $\|\mathcal{N}\|_F^2$ to suppress the confusion of noise and anomalies [43]. Finally, the ADMM [71] strategy is used to update the relevant variables, and the anomaly detection plot is obtained by calculating the final anomaly component.

A. Problem Formulation

In the TR-based method, for a given HSI $\mathcal{X} \in \mathbb{R}^{h\times w\times b}$, it is generally decomposed into a background component $\mathcal{Z} \in \mathbb{R}^{h\times w\times b}$ and an anomaly component $\mathcal{E} \in \mathbb{R}^{h\times w\times b}$, where h, w, and b denote the height, width, and number of bands of the HSI, respectively. However, due to limitations of imaging conditions or equipment, the acquired HSI is inevitably destroyed by noise, and the presence of noise brings

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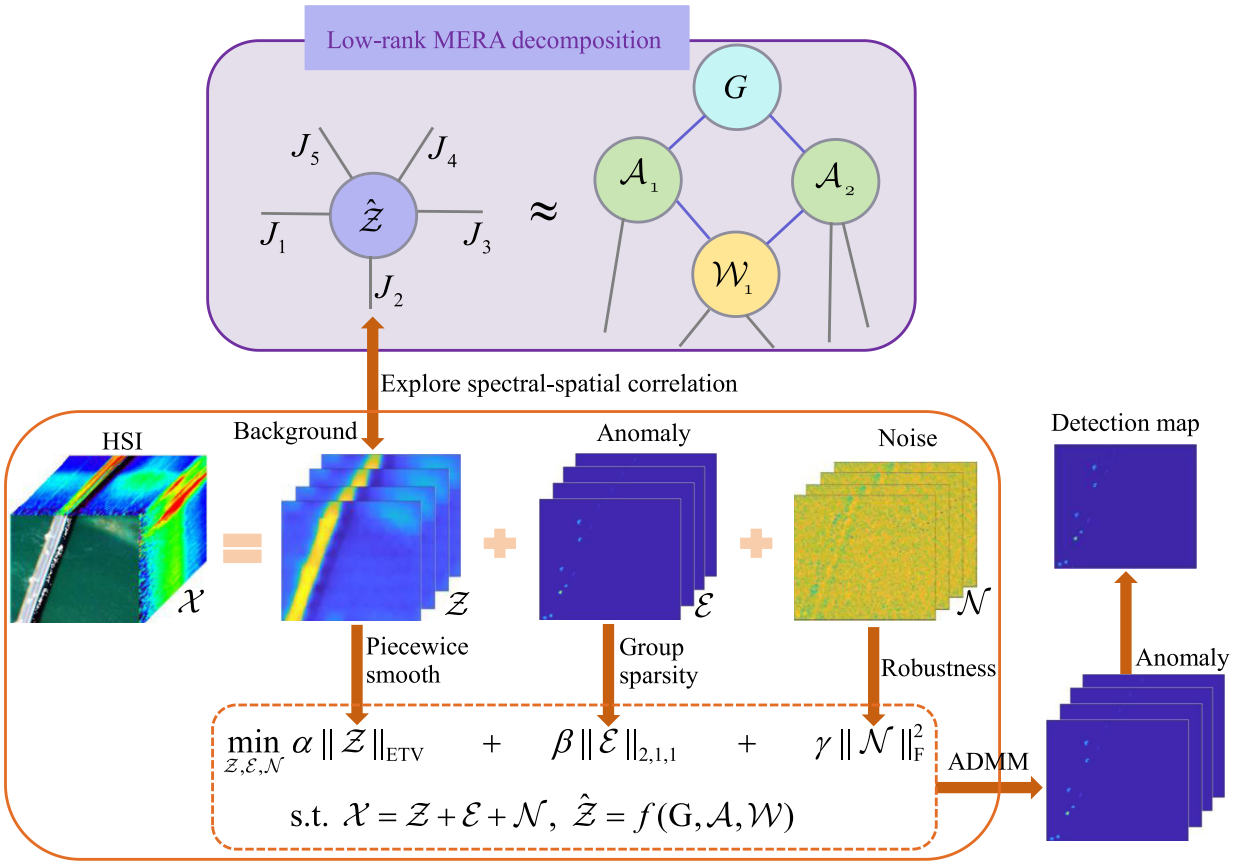


Fig. 5. Implementation flowchart of the proposed MERAETV algorithm.

challenges to anomaly detection. Therefore, in order to prevent noise from interfering with anomalies, we further decompose an additive Gaussian noise term $\mathcal{N} \in \mathbb{R}^{h\times w\times b}$ as follows:

$$\mathcal{X} = \mathcal{Z} + \mathcal{E} + \mathcal{N}. \tag{3}$$

The problem of HAD can be effectively approached as a binary classification task, where the utilization of background prior knowledge emerges as a pivotal technique to address such challenges. Low-rank and piecewise-smooth are two frequently used background priors. To mine the global low-rank of the background, the tensor decomposition methods used for HAD mainly include CP decomposition, Tucker decomposition, t-SVD, and TRD. Experiments demonstrate that these tensor decomposition methods are limited in capturing the spectral–spatial correlation of complex background components [28]. MERA decomposition is an advanced tensor network decomposition technique that decomposes a high-order tensor into several orthogonal, semiorthogonal factors, and a top-level core factor. By incorporating interactions between orthogonal and semiorthogonal (low-rank) factors, the low-rank MERA exhibits a robust representation capability, enabling it to effectively extract deep features from data and adapt to scene variability [32]. Inspired by this, the HAD model based on MERA decomposition is constructed as follows:

$$\mathcal{X} = \mathcal{Z} + \mathcal{E} + \mathcal{N}, \quad \hat{\mathcal{Z}} = f(\boldsymbol{G}, \boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{W}}) \tag{4}$$

where $\mathcal{Z} \in \mathbb{R}^{h\times w\times b}$ is reshaped into a fifth-order tensor $\hat{\mathcal{Z}} \in \mathbb{R}^{n_1\times n_2\times n_3\times n_4\times b}$ with $h = n_1 n_2$ and $w = n_3 n_4$, that is, $\hat{\mathcal{Z}} = \text{reshape}(\mathcal{Z}, [n_1, n_2, n_3, n_4, b])$. Here, n_1 and n_2 are as close to equilibrium as possible, and similarly, n_3 and n_4 are as close to equilibrium as possible. $\mathfrak{L}^{-1}$ is used to represent the reverse process of $\hat{\mathcal{Z}}$, i.e., $\mathfrak{L}^{-1}(\hat{\mathcal{Z}}) = \mathcal{Z}$. $\hat{\mathcal{Z}} = f(\boldsymbol{G}, \boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{W}})$ means applying low-rank MERA approximation on the background tensor $\hat{\mathcal{Z}}$ to capture the spectral–spatial correlation of the background. A detailed description of the solution for the low-rank MERA factors of a fifth-order tensor will be provided in Section IV-B.

As shown in Fig. 2, HSI data exhibit similar piecewise-smooth in both spectral and spatial domains. To fully utilize the prior of piecewise-smooth in the background, an ETV regularization is designed as follows:

$$\begin{aligned}\|\mathcal{Z}\|_{\text{ETV}} &= \|\nabla(\mathcal{Z})\|_1 \\ &= \|\nabla_h(\mathcal{Z})\|_1 + \|\nabla_v(\mathcal{Z})\|_1 + \|\nabla_s(\mathcal{Z})\|_1\end{aligned} \tag{5}$$

where $\nabla(\cdot) = [\nabla_h(\cdot); \nabla_v(\cdot); \nabla_s(\cdot)]$ is a 3-D difference operator. ∇_h, ∇_v, and ∇_s are first-order finite difference operators of the background tensor along the horizontal, vertical, and spectral directions, respectively, defined as $\nabla_h(\mathcal{Z}) = \mathcal{Z}_{i+1,j,k} - \mathcal{Z}_{i,j,k}$, $\nabla_v(\mathcal{Z}) = \mathcal{Z}_{i,j+1,k} - \mathcal{Z}_{i,j,k}$, $\nabla_s(\mathcal{Z}) = \mathcal{Z}_{i,j,k+1} - \mathcal{Z}_{i,j,k}$, where $i = 1, \ldots, h-1$, $j = 1, \ldots, w-1$, $k = 1, \ldots, b-1$.

Combining (4) and (5), the following model can be obtained:

$$\begin{aligned}&\min_{\mathcal{Z}} \alpha\|\mathcal{Z}\|_{\text{ETV}} \\ &\text{s.t. } \mathcal{X} = \mathcal{Z} + \mathcal{E} + \mathcal{N}, \quad \hat{\mathcal{Z}} = f(\boldsymbol{G}, \boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{W}}).\end{aligned} \tag{6}$$

In addition, an $\ell_{2,1,1}$-norm constraint is applied to the abnormal component $\mathcal{E}$ to characterize the group sparsity of

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anomalous targets, where $\|\mathcal{E}\|_{2,1,1} = \sum_{i=1}^{h}\sum_{j=1}^{w}\|\mathcal{E}(i,j,:)\|_2$. For the noise term, it is modeled as $\|\mathcal{N}\|_F^2$ to prevent confusion with anomalous targets. Based on the above analysis, the proposed MERAETV algorithm can be described as

$$\begin{aligned}&\min_{\mathcal{Z},\mathcal{E},\mathcal{N}} \alpha\|\mathcal{Z}\|_{\mathrm{ETV}} + \beta\|\mathcal{E}\|_{2,1,1} + \gamma\|\mathcal{N}\|_F^2\\ &\text{s.t. } \mathcal{X} = \mathcal{Z} + \mathcal{E} + \mathcal{N}, \quad \hat{\mathcal{Z}} = f(\boldsymbol{G}, \boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{W}})\end{aligned} \tag{7}$$

where α, β, and γ are regularization parameters. Subsequently, the detailed solution process of the MERAETV model will be shown in Section IV-C.

B. Low-Rank MERA Approximation

The effectiveness of low-rank tensor networks in approximating high-order tensor data has been demonstrated in the literature [30], [72]. The low-rank MERA approximation of a fifth-order tensor $\mathcal{Q} \in \mathbb{R}^{J_1\times\cdots\times J_5}$ aims to identify the MERA factors with predefined MERA ranks $R_1 = R_2 = R$, which is formulated as follows:

$$\min_{\boldsymbol{\mathcal{W}},\boldsymbol{\mathcal{A}},\boldsymbol{G}} \frac{1}{2}\|\mathcal{Q} - f(\boldsymbol{G}, \boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{W}})\|_F^2. \tag{8}$$

The MERA decomposition of a fifth-order tensor is illustrated in Fig. 1, revealing the presence of four distinct MERA factors: two isometries $\boldsymbol{\mathcal{A}}_1, \boldsymbol{\mathcal{A}}_2$, a disentangler $\boldsymbol{\mathcal{W}}_1$, and a top-level core $\boldsymbol{G}$. When three of them are fixed, each variable can be updated alternately.

1) *Update $\boldsymbol{\mathcal{W}}_1$:* Fixing $\boldsymbol{\mathcal{A}}_1$, $\boldsymbol{\mathcal{A}}_2$, and $\boldsymbol{G}$, the subproblem of $\boldsymbol{\mathcal{W}}_1$ will be rewritten in matrix form as follows:

$$\min_{\boldsymbol{W}_1} \frac{1}{2}\|\boldsymbol{Q}_{(2,3)} - \boldsymbol{W}_1\boldsymbol{T}_w\|_F^2, \quad \text{s.t. } \boldsymbol{W}_1^*\boldsymbol{W}_1 = \boldsymbol{I}_{J_2J_3} \tag{9}$$

where $\boldsymbol{T}_w$ and $\boldsymbol{Q}_{(2,3)}$ denote the mode-$\{2,3\}$ unfolding matrices of $\mathcal{T}_w$ and $\mathcal{Q}$, respectively. $\mathcal{T}_w$ represents the tensor contraction outcome of other variables, that is, $\mathcal{T}_w = \boldsymbol{G} \times_{\{R_1\}} \boldsymbol{\mathcal{A}}_1 \times_{\{R_2\}} \boldsymbol{\mathcal{A}}_2 \in \mathbb{R}^{J_1\times J_2\times J_3\times J_4\times J_5}$. $\boldsymbol{W}_1$ denotes the mode-$\{1,2\}$ unfolding matrix of $\boldsymbol{\mathcal{W}}_1 \in \mathbb{R}^{J_2\times J_3\times J_2\times J_3}$.

The problem of solving (9) can be reformulated as an equivalent orthogonal Procrustes problem

$$\max_{\boldsymbol{W}_1} \mathrm{Tr}((\boldsymbol{W}_1)^*\boldsymbol{Q}_{(2,3)}(\boldsymbol{T}_w)^*). \tag{10}$$

Therefore, $\boldsymbol{\mathcal{W}}_1$ can be updated by the following formula:

$$\boldsymbol{\mathcal{W}}_1 = \mathrm{fold}_{(1,2)}\left(\boldsymbol{U}_w\boldsymbol{V}_w^*\right) \tag{11}$$

where $\boldsymbol{U}_w$ and $\boldsymbol{V}_w$ denote the matrices of left and right singular vectors for matrix $\boldsymbol{Q}_{(2,3)}(\boldsymbol{T}_w)^*$, respectively.

2) *Update $\boldsymbol{\mathcal{A}}_1$, $\boldsymbol{\mathcal{A}}_2$:* Similar to solving $\boldsymbol{\mathcal{W}}_1$, update $\boldsymbol{\mathcal{A}}_i$, $i = 1, 2$ by the following formula:

$$\boldsymbol{\mathcal{A}}_i = \underset{\boldsymbol{\mathcal{A}}_i:\boldsymbol{A}_i^*\boldsymbol{A}_i=\boldsymbol{I}_{R_i}}{\arg\min} \frac{1}{2}\|\mathcal{Q} - f(\boldsymbol{G}, \boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{W}})\|_F^2. \tag{12}$$

3) *Update $\boldsymbol{G}$:* The subproblem related to $\boldsymbol{G}$ is given as follows:

$$\min_{\boldsymbol{G}} \frac{1}{2}\|\mathcal{Q} - \mathcal{D} \times_{\{R_1,R_2\}} \boldsymbol{G}\|_F^2 \tag{13}$$

where $\mathcal{D}$ represents a seventh-order layer (ref. Definition 3). With the help of the semiorthogonal property of isometries $\boldsymbol{\mathcal{A}}_i$, $i = 1, 2$, and the orthogonal property of disentangler $\boldsymbol{\mathcal{W}}_1$, the following update formula for $\boldsymbol{G}$ can be obtained:

$$\boldsymbol{G} = \mathcal{Q} \times_{\{J_1,J_2,J_3,J_4,J_5\}} \mathcal{D}. \tag{14}$$

The entire solution process of the low-rank MERA approximation is outlined in Algorithm 1.

Algorithm 1 Low-Rank MERA Approximation

Input: 5-th order tensor $\mathcal{Q}$, MERA ranks $R_1 = R_2 = R$.
1: **Initialize:** $\boldsymbol{G}$, $\boldsymbol{\mathcal{A}}$, $\boldsymbol{\mathcal{W}}$, $\mathrm{Iter}_{\max} = 10$;
2: **For** $k = 1 : \mathrm{Iter}_{\max}$ **do**
3: Update $\boldsymbol{\mathcal{W}}^k$ by (11);
4: Update $\boldsymbol{\mathcal{A}}^k$ by (12);
5: Update $\boldsymbol{G}^k$ by (14);
6: **End for**
7: $\hat{\mathcal{Q}} = f(\boldsymbol{G}^{\mathrm{Iter}_{\max}}, \boldsymbol{\mathcal{A}}^{\mathrm{Iter}_{\max}}, \boldsymbol{\mathcal{W}}^{\mathrm{Iter}_{\max}})$;
Output: Low-rank MERA approximation tensor $\hat{\mathcal{Q}}$.

C. Optimize MERAETV Model

Model (7) can be effectively solved by the ADMM algorithm. By introducing auxiliary variables $\mathcal{L}$, $\mathcal{Y}$, and $\mathcal{F}$, model (7) is rewritten as the following equivalent model with separable variables:

$$\begin{aligned}&\min_{\mathcal{Z},\mathcal{E},\mathcal{N},\mathcal{L},\mathcal{F},\mathcal{Y}} \alpha\|\mathcal{F}\|_1 + \beta\|\mathcal{E}\|_{2,1,1} + \gamma\|\mathcal{N}\|_F^2\\ &\text{s.t. } \mathcal{X} = \mathcal{Z} + \mathcal{E} + \mathcal{N}, \quad \mathcal{Z} = \mathcal{L}, \ \nabla(\mathcal{L}) = \mathcal{F}, \ \hat{\mathcal{Z}} = \mathcal{Y}\\ &\quad \mathcal{Y} = f(\boldsymbol{G}, \boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{W}}).\end{aligned} \tag{15}$$

Then, the augmented Lagrangian function of optimization model (15) is given as follows:

$$\begin{aligned}&\Phi(\mathcal{Z}, \mathcal{L}, \mathcal{F}, \mathcal{Y}, \mathcal{E}, \mathcal{N}, \mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3, \mathcal{T}_4, \mu)\\ &= \alpha\|\mathcal{F}\|_1 + \beta\|\mathcal{E}\|_{2,1,1} + \gamma\|\mathcal{N}\|_F^2\\ &\quad + \frac{\mu}{2}\left\|\mathcal{X} - \mathcal{Z} - \mathcal{E} - \mathcal{N} + \frac{\mathcal{T}_1}{\mu}\right\|_F^2\\ &\quad + \frac{\mu}{2}\left\|\mathcal{Z} - \mathcal{L} + \frac{\mathcal{T}_2}{\mu}\right\|_F^2 + \frac{\mu}{2}\left\|\nabla(\mathcal{L}) - \mathcal{F} + \frac{\mathcal{T}_3}{\mu}\right\|_F^2\\ &\quad + \frac{\mu}{2}\left\|\hat{\mathcal{Z}} - \mathcal{Y} + \frac{\mathcal{T}_4}{\mu}\right\|_F^2\end{aligned} \tag{16}$$

under the constraint $\mathcal{Y} = f(\boldsymbol{G}, \boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{W}})$, where $\mu > 0$ is the penalty parameter and $\mathcal{T}_i (i = 1, 2, 3, 4)$ are the Lagrange multipliers. Therefore, we can optimize (16) by alternately updating each variable while holding the other variables constant. Specifically, in the $(t+1)$th iteration, each variable in (16) is updated as follows.

1) *Update $\mathcal{Z}$:* By discarding irrelevant variables except $\mathcal{Z}$, (16) is simplified to

$$\begin{aligned}&\mathcal{Z}^{t+1}\\ &= \arg\min_{\mathcal{Z}} \frac{\mu^t}{2}\left\|\mathcal{X} - \mathcal{Z} - \mathcal{E}^t - \mathcal{N}^t + \frac{\mathcal{T}_1^t}{\mu^t}\right\|_F^2\\ &\quad + \frac{\mu^t}{2}\left\|\mathcal{Z} - \mathcal{L}^t + \frac{\mathcal{T}_2^t}{\mu^t}\right\|_F^2\end{aligned}$$

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$$+\frac{\mu^t}{2}\left\|\mathfrak{L}^{-1}(\hat{\mathcal{Z}})-\mathfrak{L}^{-1}(\mathcal{Y}^t)+\mathfrak{L}^{-1}\left(\frac{\mathcal{T}_4^t}{\mu^t}\right)\right\|_F^2$$
$$=\arg\min_{\mathcal{Z}}\ \frac{\mu^t}{2}\left\|\mathcal{Z}-\mathcal{P}_1^t\right\|_F^2+\frac{\mu^t}{2}\left\|\mathcal{Z}-\mathcal{P}_2^t\right\|_F^2$$
$$+\frac{\mu^t}{2}\left\|\mathcal{Z}-\mathcal{P}_3^t\right\|_F^2 \tag{17}$$

where $\mathcal{P}_1^t=\mathcal{X}-\mathcal{E}^t-\mathcal{N}^t+(\mathcal{T}_1^t/\mu^t)$, $\mathcal{P}_2^t=\mathcal{L}^t-(\mathcal{T}_2^t/\mu^t)$, and $\mathcal{P}_3^t=\mathfrak{L}^{-1}(\mathcal{Y}^t-(\mathcal{T}_4^t/\mu^t))$. By taking the derivative of $\mathcal{Z}$ in (17) and setting the derivative to zero, the following update formula for $\mathcal{Z}$ can be obtained:

$$\mathcal{Z}^{t+1}=\left(\mathcal{P}_1^t+\mathcal{P}_2^t+\mathcal{P}_3^t\right)/3. \tag{18}$$

2) *Update $\mathcal{L}$:* The optimization problem for $\mathcal{L}$ in (16) is given as follows:

$$\mathcal{L}^{t+1}=\arg\min_{\mathcal{L}}\ \frac{\mu^t}{2}\left\|\mathcal{Z}^{t+1}-\mathcal{L}+\frac{\mathcal{T}_2^t}{\mu^t}\right\|_F^2$$
$$+\frac{\mu^t}{2}\left\|\nabla(\mathcal{L})-\mathcal{F}^t+\frac{\mathcal{T}_3^t}{\mu^t}\right\|_F^2. \tag{19}$$

The solution to (19) can be expressed as a linear system, which is presented as follows:

$$\left(\mu^t\boldsymbol{I}+\mu^t\nabla^*\nabla\right)\mathcal{L}=\mu^t\mathcal{Z}^{t+1}+\mu^t\nabla^*\left(\mathcal{F}^t\right)+\mathcal{T}_2^t$$
$$-\nabla^*\left(\mathcal{T}_3^t\right). \tag{20}$$

Since operator $\nabla^*\nabla$ has a block-cyclic structure, it can be calculated by 3-D fast Fourier transform (FFT) [73]. Therefore, the update of $\mathcal{L}^{t+1}$ can be written as

$$\mathcal{L}^{t+1}=\text{ifftn}\left(\frac{\text{fftn}(\mathcal{M})}{\mu^t\mathbf{1}+\mu^t(\text{fftn}(\nabla^*\nabla))}\right) \tag{21}$$

where $\mathcal{M}=\mu^t\mathcal{Z}^{t+1}+\mu^t\nabla^*(\mathcal{F}^t)+\mathcal{T}_2^t-\nabla^*(\mathcal{T}_3^t)$, and fftn and ifftn are 3-D FFT and its inverse transform, respectively.

3) *Update $\mathcal{F}$:* The subproblem about $\mathcal{F}$ in (16) is given as follows:

$$\mathcal{F}^{t+1}=\arg\min_{\mathcal{F}}\ \alpha\|\mathcal{F}\|_1$$
$$+\frac{\mu^t}{2}\left\|\mathcal{F}-\left(\nabla\left(\mathcal{L}^{t+1}\right)+\frac{\mathcal{T}_3^t}{\mu^t}\right)\right\|_F^2. \tag{22}$$

Equation (22) can be effectively updated by the soft-thresholding operator

$$\mathcal{F}^{t+1}=\Theta_{\frac{\alpha}{\mu^t}}\left(\nabla\left(\mathcal{L}^{t+1}\right)+\frac{\mathcal{T}_3^t}{\mu^t}\right) \tag{23}$$

where $\Theta_\zeta(\cdot)$ represents the soft-thresholding operator as follows:

$$\Theta_\zeta(x)=\text{sign}(x)\cdot\max(|x|-\zeta,0). \tag{24}$$

4) *Update $\mathcal{Y}$:* Extracting all terms about $\mathcal{Y}$ in (16), we can get

$$\mathcal{Y}^{t+1}=\arg\min_{\mathcal{Y}}\ \frac{\mu^t}{2}\left\|\mathcal{Y}-\left(\hat{\mathcal{Z}}^{t+1}+\frac{\mathcal{T}_4^t}{\mu^t}\right)\right\|_F^2$$
$$\text{s.t. }\mathcal{Y}=f(\boldsymbol{G},\boldsymbol{\mathcal{A}},\boldsymbol{\mathcal{W}}). \tag{25}$$

Algorithm 2 MERAETV Algorithm for HAD

Input: HSI tensor data $\boldsymbol{\mathcal{X}}\in\mathbb{R}^{h\times w\times b}$, regularization parameters α, β and γ.
1: **Initialize:** $t=0$, $\mathcal{Z}^0=\mathcal{L}^0=\mathcal{E}^0=\mathcal{N}^0=\mathcal{T}_1^0=\mathcal{T}_2^0=\mathbf{0}$, $\mathcal{T}_3^0=\mathcal{F}^0=\mathbf{0}$, $\mathcal{T}_4^0=\mathcal{Y}^0=\mathbf{0}$, $\mu^0=1e$-5, $\delta=1e$-4, $\rho=1.5$, $\mu_{\max}=1e10$, and max iteration number $\varphi=50$;
2: **While** ($0\le t\le\varphi$ or not converged) **do**;
3: Update $\mathcal{Z}^{t+1}$ via (18);
4: Update $\mathcal{L}^{t+1}$ via (21);
5: Update $\mathcal{F}^{t+1}$ via (23);
6: Update $\mathcal{Y}^{t+1}$ via (25);
7: Update $\mathcal{E}^{t+1}$ via (27);
8: Update $\mathcal{N}^{t+1}$ via (28);
9: Update the Lagrangian multipliers $\mathcal{T}_i^{t+1}(i=1,2,3,4)$ and penalty parameter μ^{t+1} via (29);
10: Update iteration: $t=t+1$;
11: The convergence condition is satisfied via (30);
12: **End while**
13: Get the optimal abnormal tensor $\mathcal{E}_*$;
14: Anomaly detection map $\boldsymbol{T}$ is obtained by $\boldsymbol{T}_{ij}=\sqrt{\sum_{k=1}^b|\mathcal{E}_*(i,j,k)|^2}$;
Output: Anomaly detection map $\boldsymbol{T}$.

Let $\mathcal{P}_4^t=\hat{\mathcal{Z}}^{t+1}+(\mathcal{T}_4^t/\mu^t)$; by using $\mathcal{P}_4^t$ as the input data of Algorithm 1, the solution of (25) can be obtained.

5) *Update $\mathcal{E}$:* Regarding $\mathcal{E}$, (16) is rewritten as follows:

$$\mathcal{E}^{t+1}=\arg\min_{\mathcal{E}}\ \beta\|\mathcal{E}\|_{2,1,1}$$
$$+\frac{\mu^t}{2}\left\|\mathcal{X}-\mathcal{Z}^{t+1}-\mathcal{E}-\mathcal{N}^t+\frac{\mathcal{T}_1^t}{\mu^t}\right\|_F^2. \tag{26}$$

Let $\mathcal{P}_5^t=\mathcal{X}-\mathcal{Z}^{t+1}-\mathcal{N}^t+(\mathcal{T}_1^t/\mu^t)$: the closed-form solution of (26) can be achieved

$$\mathcal{E}^{t+1}(i,j,:)=\left(1-\frac{\beta}{\mu^t\|\mathcal{P}_5^t(i,j,:)\|_2}\right)_+*\mathcal{P}_5^t(i,j,:) \tag{27}$$

where $i\in\{1,\ldots,h\}$, $j\in\{1,\ldots,w\}$ and $\zeta_+=\max(\zeta,0)$.

6) *Update $\mathcal{N}$:* The optimization subproblem for $\mathcal{N}$ is given as follows:

$$\mathcal{N}^{t+1}=\arg\min_{\mathcal{N}}\ \gamma\|\mathcal{N}\|_F^2$$
$$+\frac{\mu^t}{2}\left\|\mathcal{N}-\left(\mathcal{X}-\mathcal{Z}^{t+1}-\mathcal{E}^{t+1}+\frac{\mathcal{T}_1^t}{\mu^t}\right)\right\|_F^2$$
$$=\frac{\mu^t(\mathcal{X}-\mathcal{Z}^{t+1}-\mathcal{E}^{t+1})+\mathcal{T}_1^t}{\mu^t+2\gamma}. \tag{28}$$

7) *Update the Lagrange Multipliers $\mathcal{T}_i(i=1,2,3,4)$ and Penalty Parameter μ:*

$$\begin{cases}\mathcal{T}_1^{t+1}=\mathcal{T}_1^t+\mu^t(\mathcal{X}-\mathcal{Z}^{t+1}-\mathcal{E}^{t+1}-\mathcal{N}^{t+1})\\ \mathcal{T}_2^{t+1}=\mathcal{T}_2^t+\mu^t(\mathcal{Z}^{t+1}-\mathcal{L}^{t+1})\\ \mathcal{T}_3^{t+1}=\mathcal{T}_3^t+\mu^t(\nabla(\mathcal{L}^{t+1})-\mathcal{F}^{t+1})\\ \mathcal{T}_4^{t+1}=\mathcal{T}_4^t+\mu^t(\hat{\mathcal{Z}}^{t+1}-\mathcal{Y}^{t+1})\\ \mu^{t+1}=\min(\rho\mu^t,\mu_{\max}).\end{cases} \tag{29}$$

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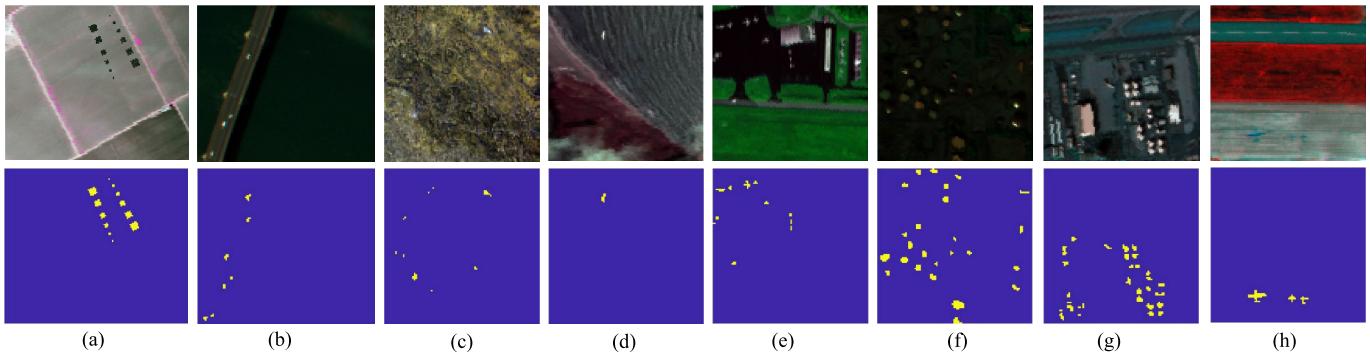


Fig. 6. False color images and corresponding ground-truth maps. (a) Salinas. (b) Pavia. (c) Hyperion. (d) Beach-3. (e) Urban-3. (f) Urban-4. (g) Urban-5. (h) Airport-4.

TABLE I
DETAILS OF EIGHT DATASETS

Datasets	Image size	Location	Sensors	Resolution	Proportion
Salinas	126×150×204	Salinas	AVIRIS	3.7m	0.98%
Pavia	105×120×102	Pavia	ROSIS	1.3m	0.34%
Hyperion	100×100×145	Okavango	Hyperion	30m	0.32%
Beach-3	100×100×188	Champagne	AVIRIS	4.4m	0.13%
Urban-3	100×100×191	Gainesville	AVIRIS	3.5m	0.52%
Urban-4	100×100×205	Los Angeles	AVIRIS	7.1m	1.70%
Urban-5	100×100×205	Los Angeles	AVIRIS	7.1m	2.32%
Airport-4	100×100×191	Gulfport	AVIRIS	3.4m	0.60%

8) The condition for convergence can be stated as

$$\text{Error} = \frac{\|\boldsymbol{\mathcal{X}} - \boldsymbol{\mathcal{Z}}^{t+1} - \boldsymbol{\mathcal{E}}^{t+1} - \boldsymbol{\mathcal{N}}^{t+1}\|_F}{\|\boldsymbol{\mathcal{X}}\|_F} \leq \delta. \quad (30)$$

The detailed solution process of MERAETV is shown in Algorithm 2.

V. EXPERIMENT ANALYSIS

In this section, we conduct experiments on one synthetic and seven real HSI datasets to evaluate the performance of the proposed MERAETV algorithm. It is worth noting that all data need to be normalized band by band before executing the algorithm. Furthermore, all numerical results are achieved in MATLAB 2022a on the computer with Intel Core i7-12700F processors, 2.10-GHz CPU, and 32 GB of memory.

A. Datasets' Description

The first Salinas is a synthetic dataset offered by Verdoja and Grangetto [74]. This synthetic dataset is generated based on the Salinas real HSI dataset. The spatial size of the Salinas real HSI dataset is 512 × 217 and contains 224 bands. After removing the low signal-to-noise ratio and water vapor absorption bands, 204 effective bands are retained for experiments. In the experiment, the synthetic abnormal targets are achieved by using the target implantation approach [75] in different parts of the Salinas real scene. The 126 × 150 ground-truth image $\boldsymbol{U}$ displayed in Fig. 6(a) is achieved by constructing six squares with side lengths of 1–6 pixels and arranging them in a row. The six squares are then copied in reverse order and arranged closely into another row. Among them, the rotation angle of these two rows is approximately $\pi/6$. Except for the pixel values within the 12 squares, which are 1, the other pixel values in $\boldsymbol{U}$ are all 0. Finally, in the Salinas real scene, an area $\boldsymbol{J}$ that is the same size as the ground-truth image is cropped as the background image. By using $\boldsymbol{J}$ to construct the synthetic data $\acute{\boldsymbol{J}}$ containing the implanted target, the expression is given as follows:

$$\acute{\boldsymbol{J}}_{ij} = \boldsymbol{U}_{ij} \cdot \Psi(g) + (1 - \boldsymbol{U}_{ij}) \cdot \boldsymbol{J}_{ij} \quad (31)$$

where $\Psi(\cdot)$ is a function representing the target spectral abundance and g is determined according to the target category. The pseudocolor image and the corresponding ground-truth image of the synthesized Salinas dataset are shown in Fig. 6(a).

The second Pavia dataset is captured by the Reflection Optical System Imaging Spectroradiometer (ROSIS) above the center of Pavia, Italy. The background mainly consists of rivers, bridges, and soil, with vehicles on bridges considered anomaly targets [76]. The third Hyperion dataset is taken by the Hyperion sensor carried by EO-1 in the Okavango Delta in 2001 [77]. The remaining five datasets are from the Airport-Beach-Urban (ABU)[1] dataset. The details of the above eight datasets are listed in Table I, while Fig. 6 shows the false color maps and corresponding ground-truth maps of these datasets.

B. Comparison Algorithms

In order to verify the effectiveness of the proposed MERAETV algorithm, 11 representative HAD algorithms are selected as comparison algorithms, namely, LRX [5], FrFE [7], CRD [11], RPCA [14], GTVLRR [51], LSDM-MoG [61], FEBPAD [63], GAED [8], PCA-TLRSR [44], PTA [24], and LARTVAD [25]. Among them, LRX and FrFE are algorithms based on statistical theory; CRD is an algorithm based on CR; RPCA, LSDM-MoG, and FEBPAD are algorithms based on LRaSMD; GTVLRR is an algorithm based on LRaSR; GAED is an algorithm based on deep learning; and PCA-TLRSR, PTA, and LARTVAD are the three state-of-the-art algorithms based on TR.

C. Experimental Settings

Regarding the parameter value issues involved in all comparison algorithms, we adopt the value range recommended by the author of the corresponding article and fine-tune it to report its best performance. The specific parameter values of various comparison algorithms on different datasets are exhibited in Table II. LRX involves two parameters, namely, the inner window w_{in} and the outer window w_{out}. These

[1] http://xudongkang.weebly.com/

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TABLE II
PARAMETER VALUES OF EACH ALGORITHM ON DIFFERENT DATASETS

Methods	Parameters	Salinas	Pavia	Hyperion	Beach-3	Urban-3	Urban-4	Urban-5	Airport-4
LRX	w_{in}, w_{out}	(5,7)	(7,9)	(5,11)	(11,13)	(5,11)	(3,5)	(3,5)	(3,5)
CRD	w_{in}, w_{out}	(9,11)	(11,13)	(13,15)	(3,5)	(17,19)	(9,11)	(17,19)	(11,13)
	λ	1e-6	1e-6	1e-6	1e-6	1e-6	1e-6	1e-6	1e-6
RPCA	λ	1e-2	1e-3	1e-3	1e-2	1e-2	1e-2	1e-2	1e-2
GTVLRR	K, P, λ	15,20,0.5	15,20,0.5	15,20,10	15,20,0.01	15,20,0.5	15,20,10	15,20,0.5	15,20,0.5
	β, γ	0.2,0.05	0.2,0.05	0.05,0.01	0.005,0.01	0.2,0.05	0.5,0.01	0.2,0.05	0.2,0.05
LSDM-MoG	l_0, k	10,2	10,2	10,2	20,2	10,2	10,2	10,2	10,2
FEBPAD	r, p	3,0.4	3,0.5	3,0.8	3,0.7	3,0.5	3,0.7	3,0.7	3,0.7
	k	$0.7 \times h \times w$	$0.7 \times h \times w$	$0.7 \times h \times w$	$0.7 \times h \times w$	$0.7 \times h \times w$	$0.7 \times h \times w$	$0.7 \times h \times w$	$0.7 \times h \times w$
GAED	la, r_{iter}, β	0.1,500,1	0.4,500,1	0.1,300,1	0.1,300,1	0.1,300,1	0.4,500,10	0.1,300,1	0.1,300,1
	c, n_{hid}	7,25	7,25	7,25	7,25	7,25	7,25	7,25	7,25
PCA-TLRSR	d, λ_1	6,0.02	17,0.06	4,0.06	15,0.06	4,0.06	5,0.02	4,0.06	17,0.06
	λ_2, ω	0.01,1	0.05,5	0.01,5	0.01,5	0.01,5	0.05,10	0.01,5	0.05,1
PTA	r, α, β	1,1e-2,1e-3	1,1,1e-2	1,1e-2,1e-2	1,1e-3,1e-2	1,1e-4,1	1,1e-3,1e-3	1,1e-2,1e-2	1,1e-4,1
	τ, μ	1,1e-4	1,1e-4	1e-2,1e-4	1,1e-4	1,1e-4	1,1e-4	1e-2,1e-4	1,1e-4
LARTVAD	K, β, λ	80,0.5,0.03	80,10,2	100,0.5,15	80,0.5,3	100,2,5	80,0.5,0.1	100,2,5	80,10,2
Ours	R, α	2,1	2,0.01	12,1e-5	26,0.01	12,0.001	8,0.5	16,0.01	2,0.01
	β, γ	0.01,0.01	0.05,0.025	0.01,0.1	0.06,0.01	0.05,0.05	0.06,10	0.09,0.1	0.05,0.01

are three parameters in CRD, namely, w_{in}, w_{out}, and the tradeoff parameter λ. RPCA has only one tuning parameter λ. There are five parameters in GTVLRR, namely, the number of clusters K, the number of atoms selected in each cluster P, and the three regularization parameters λ, β, γ. For LSDM-MoG, there is only an initialization rank l_0 and the number of noises k. For FEBPAD, it involves the fractional transformation order p, the rank r of the low-rank matrix, and the cardinality k of the sparse anomaly matrix. GAED contains five parameters: learning rate la, number of iterations r_{iter}, penalty coefficient β, window size c, and hidden layer dimension n_{hid}. PCA-TLRSR has four parameters: PCA preprocessing dimension d, weight coefficient ω, and regularization parameters λ_1 and λ_2. PTA involves five parameters: truncated low-rank r, regularization parameters α and β, and penalty coefficients τ and μ. For LARTVAD, it includes the number of local clusters K and two regularization parameters β and λ. The parameters involved in the MERAETV algorithm proposed in this article will be analyzed in detail in Section V-F.

D. Evaluation Metrics

In our experiments, four commonly used evaluation metrics are adopted, namely, anomaly detection map, 3-D receiver operating characteristic (ROC) curve [78], area under the ROC curve (AUC), and box-whisker map [79]. 3-D ROC describes the relationship between detection probability P_D, false alarm probability P_F, and detection threshold τ. Moreover, three 2-D ROC curves can be decomposed from the 3-D ROC curve, namely, (P_D, P_F), (P_D, τ), and (P_F, τ). Correspondingly, we can get $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$. By combining these three evaluation indicators with each other, the following five comprehensive evaluation indicators can be obtained and defined as:

$$\begin{cases} \text{AUC}_{\text{TD}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} \\ \text{AUC}_{\text{BS}} = \text{AUC}_{(D,F)} - \text{AUC}_{(F,\tau)} \\ \text{AUC}_{\text{SNPR}} = \dfrac{\text{AUC}_{(D,\tau)}}{\text{AUC}_{(F,\tau)}} \\ \text{AUC}_{\text{TDBS}} = \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \\ \text{AUC}_{\text{ODP}} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \end{cases} \tag{32}$$

where AUC_{TD} evaluates the joint detection capability of the detector, AUC_{BS} evaluates the joint background suppression capability of the detector, AUC_{SNPR} evaluates the signal-to-noise ratio of the detector, AUC_{TDBS} evaluates the comprehensive target detection and background suppression capabilities of the detector, and AUC_{ODP} evaluates the overall detection probability of the detector.

In general, $\text{AUC}_{(D,F)}$, AUC_{TDBS}, and AUC_{ODP} are used to evaluate the effectiveness of the detector. $\text{AUC}_{(D,\tau)}$ and AUC_{TD} are used to evaluate the target detection capability of the detector. $\text{AUC}_{(F,\tau)}$, AUC_{BS}, and AUC_{SNPR} are used to evaluate the background suppression ability of the detector. In theory, an excellent detector should have very small $\text{AUC}_{(F,\tau)}$ and very large other AUC values. The box-whisker map is primarily utilized for assessing the discriminability between anomalies and background. The greater the gap between the anomaly box and the background box, the enhanced separation and detection effect of the detector can be achieved.

E. Detection Performance

1) Qualitative Analysis: The anomaly detection plots of different detectors on the Salinas dataset are exhibited in Fig. 7(a). It can be easily found that the PTA detector mistakenly identifies more backgrounds as abnormal targets, resulting in the worst visual effect, followed by PCA-TLRSR and GTVLRR. The GAED, FEBPAD, RPCA, and FrFE detectors have better background suppression, but abnormal targets can hardly be detected, especially the RPCA detector. The LARTVAD, LSDM-MoG, CRD, and LRX detectors can observe the approximate distribution of abnormal targets but suffer from severe background contamination. Compared with other detectors, the proposed MERAETV detector achieves a detection result that is almost identical to the ground truth. For the Pavia dataset, as we observe in Fig. 7(b), all detectors except LRX can detect abnormal targets. Compared with all competitors, the proposed MERAETV detector obtains the visually best anomaly detection map. The anomaly detection plots on the Hyperion dataset are shown in Fig. 7(c). It can be observed that the visual effects of the anomaly detection maps

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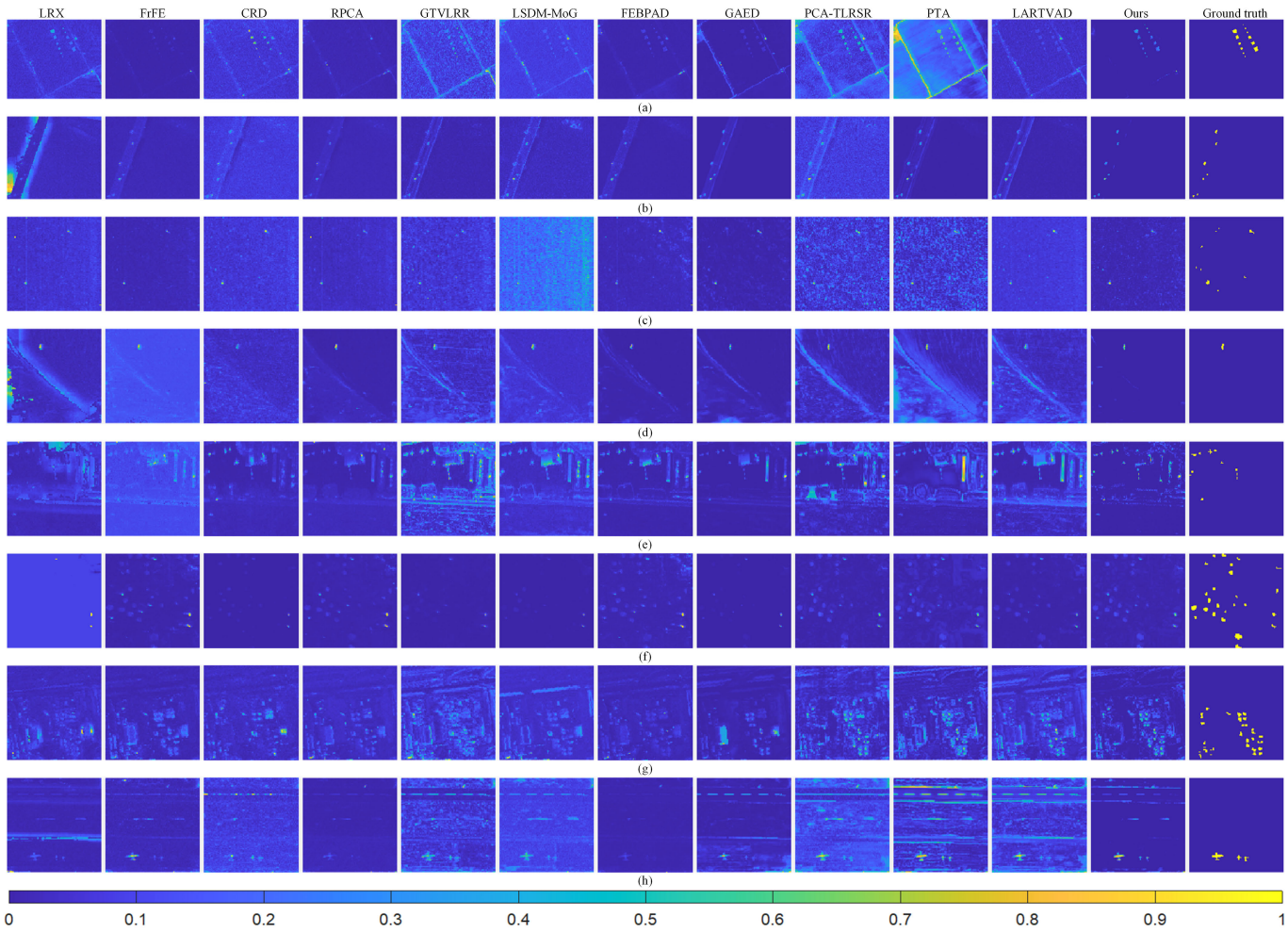


Fig. 7. Anomaly detection maps obtained on (a) Salinas, (b) Pavia, (c) Hyperion, (d) Beach-3, (e) Urban-3, (f) Urban-4, (g) Urban-5, and (h) Airport-4.

obtained by LSDM-MoG, PCA-TLRSR, and PTA detectors are the worst. The LARTVAD, GTVLRR, CRD, and LRX seem to achieve similar anomaly detection results but suffer from heavier background contamination. In comparison, the proposed MERAETV, GAED, FEBPAD, RPCA, and FrFE detectors have better background suppression, among which MERAETV can well locate and identify abnormal targets. As shown in the anomaly detection plots in Fig. 7(d), on the Beach-3 dataset, each detector can detect the abnormal target position well, but LARTVAD, PTA, PCA-TLRSR, LSDM-MoG, GTVLRR, and LRX have higher false alarm rates. On the contrary, the proposed MERAETV detector achieves a satisfactory visual detection effect.

For the Urban-3 dataset exhibited in Fig. 7(e), in an urban scene with a relatively complex background, PTA, GAED, FEBPAD, RPCA, CRD, FrFE, and LRX detectors miss detection more seriously. Although the general distribution of each target can be observed, the specific shape cannot be detected and the edge contours are blurred. The GTVLRR detector generates a large number of false alarms, causing detection failure. The PCA-TLRSR, LARTVAD, and LSDM-MoG can locate abnormal target locations but are subject to heavy background contamination. On the contrary, the proposed MERAETV detector introduces fewer false alarms and has strong target detection performance. For the urban-4 dataset presented in Fig. 7(f), the proposed MERAETV, PTA, PCA-TLRSR, and FEBPAD seem to achieve similar anomaly detection effects, outperforming other detectors. Compared with other detectors, GAED, LSDM-MoG, GTVLRR, and CRD achieve better background suppression effects. For the Urban-5 dataset presented in Fig. 7(g), the GAED, FEBPAD, LSDM-MoG, RPCA, CRD, FrFE, and LRX detectors have better suppression capabilities against the background, but the response to anomalies is not visually significant. The anomaly detection map obtained by the GTVLRR detector has the worst visual effect and generates a great deal of false alarms. Compared with PTA, PCA-TLRSR, and LARTVAD, the proposed MERAETV has better background suppression ability and better visual effects. For the Airport-4 dataset reported in Fig. 7(h), LARTVAD, GAED, FEBPAD, RPCA, CRD, FrFE, and LRX cannot clearly detect two small aircraft. The GTVLRR and LSDM-MoG can detect three planes but are in poor shape. The PCA-TLRSR and PTA can detect three aircraft of different sizes but are seriously affected by background interference, especially PCA-TLRSR. Compared to alternative detectors, the MERAETV has excellent background suppression and target-highlighting capabilities.

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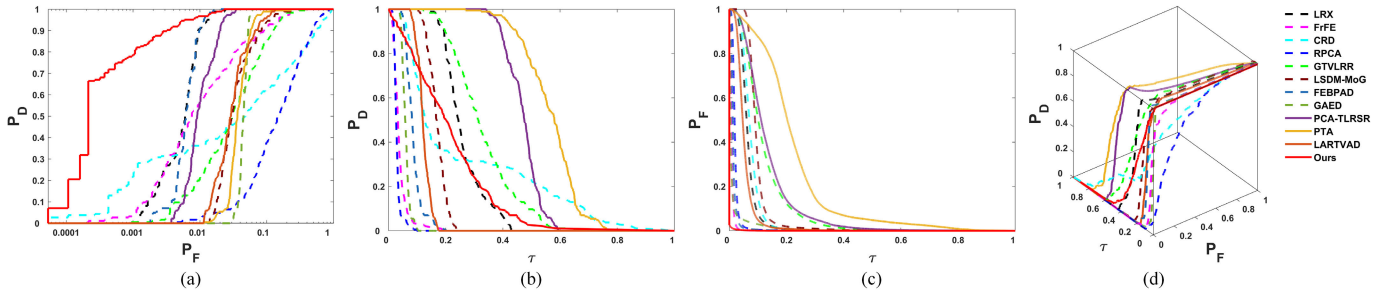


Fig. 8. ROC curves on the Salinas dataset. (a) ROC curves of (P_D, τ, P_F). (b) ROC curves of (P_D, P_F). (c) ROC curves of (P_D, τ). (d) ROC curves of (P_F, τ).

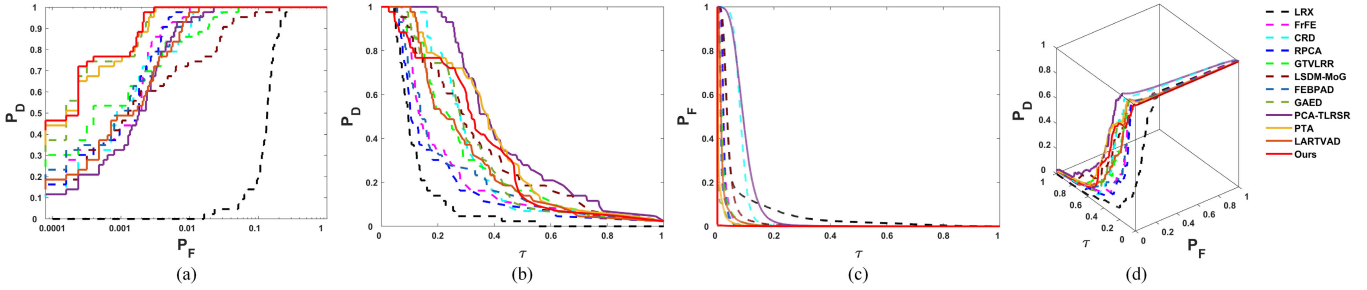


Fig. 9. ROC curves on the Pavia dataset. (a) ROC curves of (P_D, τ, P_F). (b) ROC curves of (P_D, P_F). (c) ROC curves of (P_D, τ). (d) ROC curves of (P_F, τ).

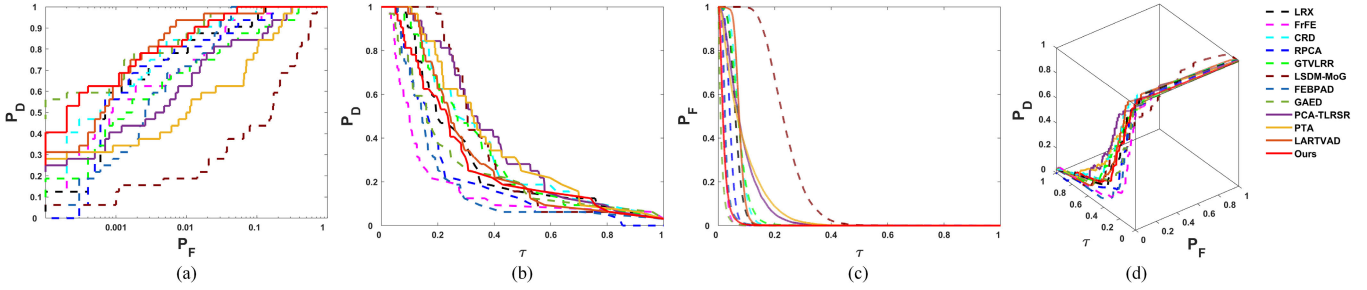


Fig. 10. ROC curves on Hyperion. (a) ROC curves of (P_D, τ, P_F). (b) ROC curves of (P_D, P_F). (c) ROC curves of (P_D, τ). (d) ROC curves of (P_F, τ).

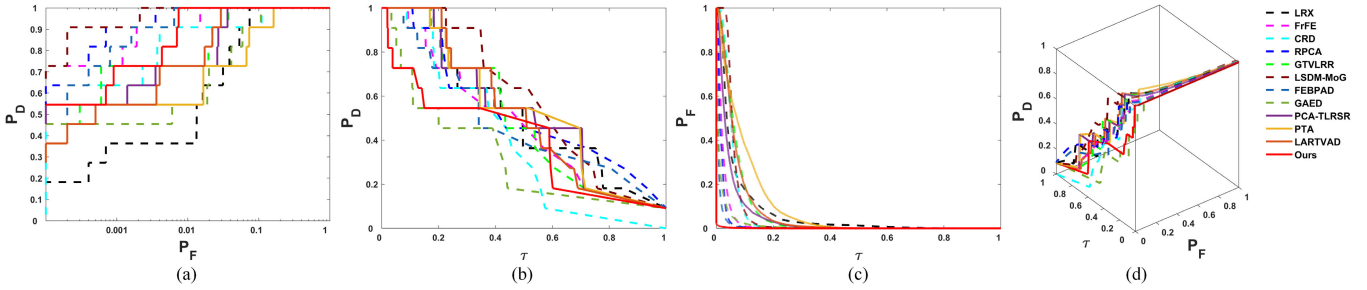


Fig. 11. ROC curves on the Beach-3 dataset. (a) ROC curves of (P_D, τ, P_F). (b) ROC curves of (P_D, P_F). (c) ROC curves of (P_D, τ). (d) ROC curves of (P_F, τ).

The 3-D ROC curves and 2-D ROC curves of each algorithm on the eight datasets are shown in Figs. 8–15. As shown in Figs. 8(a)–15(a), except for the Beach-3 dataset, the proposed MERAETV algorithm almost achieves a higher detection rate than other algorithms under different false alarm rates. On the Hyperion and Urban-4 datasets, although the ROC curves of the proposed MERAETV algorithm intersect with those of other algorithms, the ROC curves of the MERAETV algorithm are generally better than other competitors, as we can see from Figs. 10(a) and 13(a). For the Beach-3 dataset, as observed in Fig. 11(a), although the ROC curves of the LSDM-MoG, RPCA, FEBPAD, and FrFE algorithms are better than the proposed MERAETV algorithm, they show poor performance on other ROC curves. As can be observed from Figs. 8(b) to 15(b), the proposed MERAETV algorithm outperforms most of its competitors in the ROC curves on the Pavia, Urban-3, Urban-4, Urban-5, and Airport-4 datasets, exhibiting better target detection performance, especially on the Urban-4

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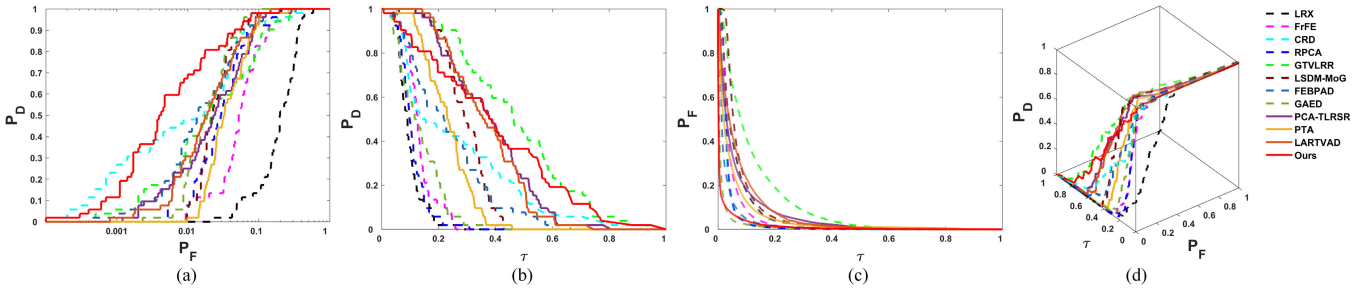


Fig. 12. ROC curves on the Urban-3 dataset. (a) ROC curves of (P_D, τ, P_F). (b) ROC curves of (P_D, P_F). (c) ROC curves of (P_D, τ). (d) ROC curves of (P_F, τ).

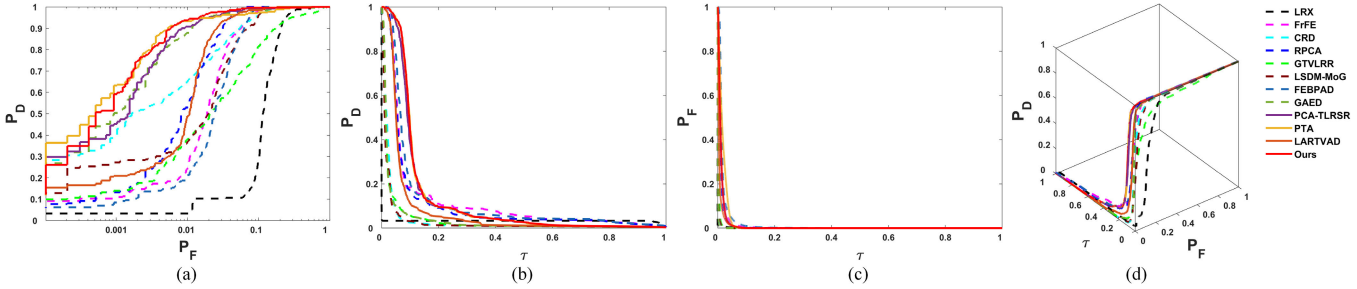


Fig. 13. ROC curves on the Urban-4 dataset. (a) ROC curves of (P_D, τ, P_F). (b) ROC curves of (P_D, P_F). (c) ROC curves of (P_D, τ). (d) ROC curves of (P_F, τ).

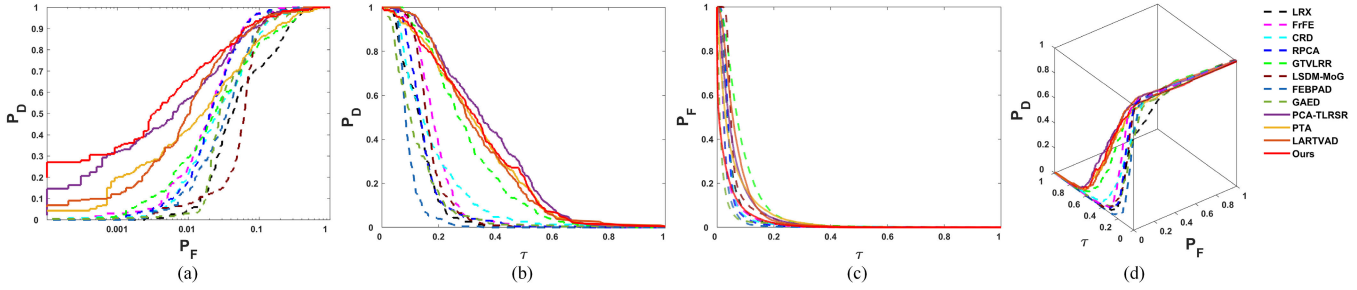


Fig. 14. ROC curves on the Urban-5 dataset. (a) ROC curves of (P_D, τ, P_F). (b) ROC curves of (P_D, P_F). (c) ROC curves of (P_D, τ). (d) ROC curves of (P_F, τ).

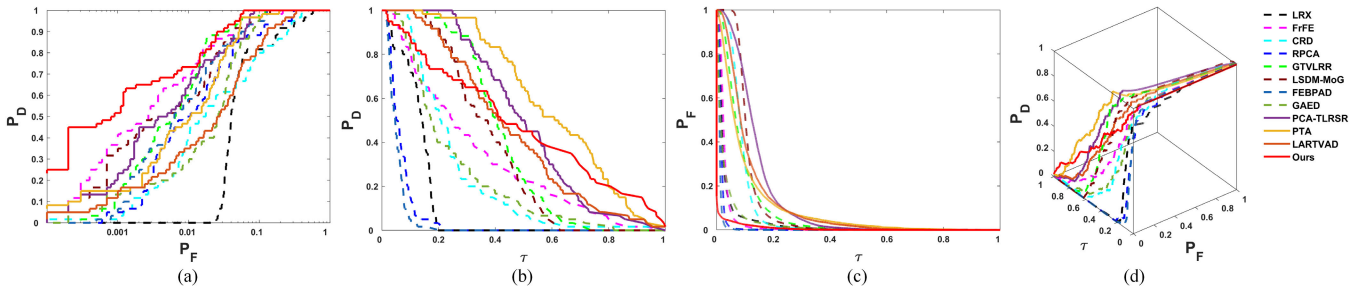


Fig. 15. ROC curves on Airport-4. (a) ROC curves of (P_D, τ, P_F). (b) ROC curves of (P_D, P_F). (c) ROC curves of (P_D, τ). (d) ROC curves of (P_F, τ).

dataset. As can be seen from Figs. 8(c) to 15(c), the ROC curves of the proposed MERAETV algorithm are almost all in the lower left corner on the eight datasets; this shows that MERAETV algorithm has a very low false alarm rate under different threshold τ's.

The box-whisker maps of different detectors on eight datasets are displayed in Fig. 16. The red box and the black box of the box-whisker plot represent the anomaly and background, respectively, and the gap between the black box and the red box represents the degree of separation between the anomaly and the background. The larger the gap, the stronger the ability to extract anomalies from the background. The observation from Fig. 16 reveals that the abnormal box of MERAETV is not always at the highest position for different datasets, but the position of the background box is almost always at the lowest position, which shows that the MERAETV method has a strong inhibitory effect on the background. It is obvious that this is consistent with the

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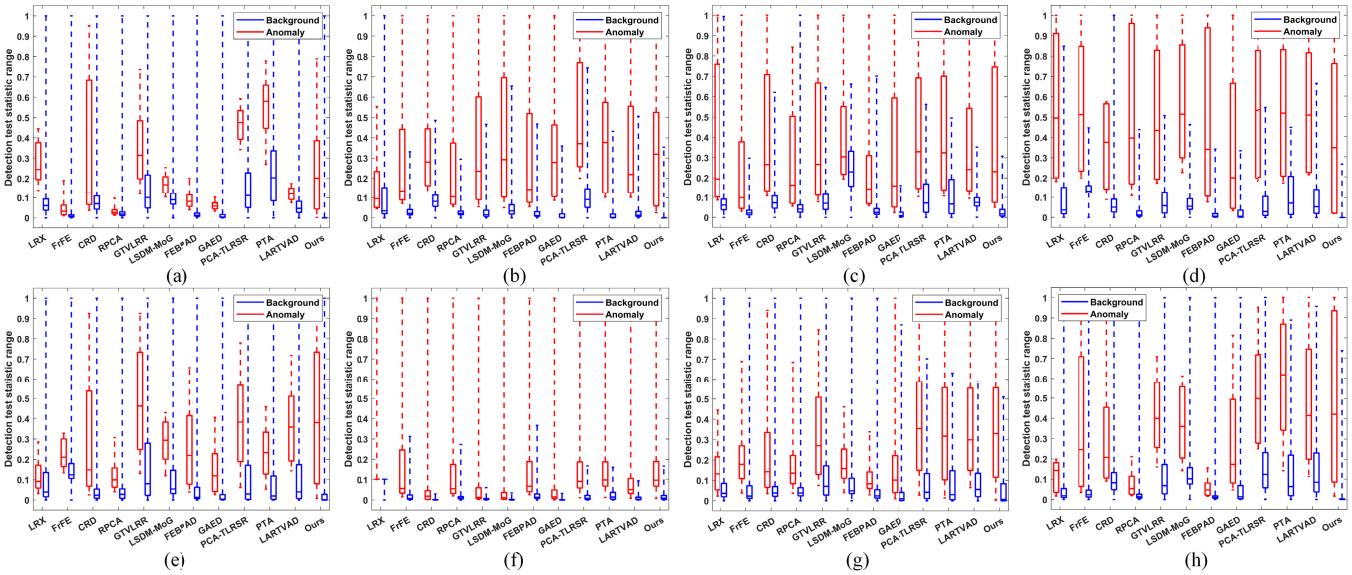


Fig. 16. Box-whisker maps for (a) Salinas, (b) Pavia, (c) Hyperion, (d) Beach-3, (e) Urban-3, (f) Urban-4, (g) Urban-5, and (h) Airport-4.

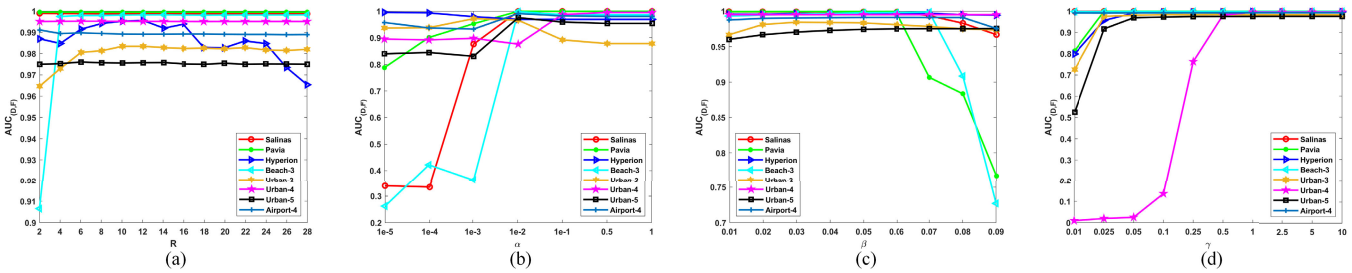


Fig. 17. Impact of different parameters on $\mathrm{AUC}_{(\mathrm{D,F})}$ of the proposed MERAETV algorithm. (a) R. (b) α. (c) β. (d) γ.

above visual analysis results, indicating that the proposed MERAETV method is able to separate abnormal targets from different background scenes.

2) Quantitative Analysis: The AUC values of different HAD algorithms on eight datasets are recorded in Table III, with optimal results highlighted in bold and suboptimal results marked with blue underline. Regarding the overall detection performance index $\mathrm{AUC}_{\mathrm{ODP}}$ of the evaluation detector, it can be easily observed from Table III that the proposed MERAETV algorithm achieves the optimal $\mathrm{AUC}_{\mathrm{ODP}}$ on the Urban-3, Urban-4, and Urban-5 datasets but achieves suboptimal $\mathrm{AUC}_{\mathrm{ODP}}$ on the Pavia, Hyperion, and Airport-4 datasets. Regarding the effectiveness metric $\mathrm{AUC}_{(D,F)}$ for evaluating the detector, except for the Beach-3 dataset, the proposed MERAETV algorithm outperforms all competing algorithms on the other seven datasets. When it comes to the background suppression performance evaluation indicators $\mathrm{AUC}_{\mathrm{BS}}$ and $\mathrm{AUC}_{\mathrm{SNPR}}$, in addition to the Urban-4 dataset, the proposed MERAETV algorithm reaches the optimal $\mathrm{AUC}_{\mathrm{BS}}$ and $\mathrm{AUC}_{\mathrm{SNPR}}$ on the other six datasets while reaching suboptimal $\mathrm{AUC}_{\mathrm{BS}}$ and $\mathrm{AUC}_{\mathrm{SNPR}}$ on the Hyerion dataset. These quantitative experimental results effectively prove the superiority of the proposed MERAETV algorithm compared to other competing algorithms.

The proposed MERAETV algorithm demonstrates distinct advantages in detecting various types of abnormal targets across diverse scenarios, as evidenced by the results obtained from both qualitative and quantitative experiments. Compared with tensor-based methods (PCA-TLRSR, PTA, and LARTVAD), the proposed MERAETV achieves optimal performance on all datasets. This may be because the proposed MERAETV incorporates MERA decomposition and ETV regularization, which fully explores the global spectral–spatial correlation and local piecewise-smooth of the background tensor, making the estimated background closer to the real background.

F. Parameter Analysis

The proposed MERAETV algorithm involves four parameters that need to be tuned, namely, the MERA rank R and three tradeoff parameters α, β, and γ. In our experiments, we tune a certain parameter by fixing other parameters. The parameter analysis details are illustrated in Fig. 17, where the horizontal and vertical coordinates represent the range of parameter values and $\mathrm{AUC}_{(D,F)}$, respectively.

1) MERA Rank R: To evaluate the influence of MERA rank R involved in MERA decomposition on detection performance, a range of numbers from 2 to 28 with an interval of 2 is selected, as depicted in Fig. 17(a). $\mathrm{AUC}_{(D,F)}$ of our MERAETV method exhibits significant fluctuations with varying R for the Hyperion dataset, while it remains relatively stable across other datasets. When $R = 12$, the proposed MERAETV can achieve optimal $\mathrm{AUC}_{(D,F)}$ values on all datasets.

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TABLE III
AUC VALUES OF VARIOUS HAD ALGORITHMS ON EIGHT HSI DATASETS

Datasets	AUC	LRX	FrFE	CRD	RPCA	GTVLRR	LSDM-MoG	FEBPAD	GAED	PCA-TLRSR	PTA	LARTVAD	Ours
Salinas	$AUC_{(D,F)}$	0.9939	0.9708	0.8341	0.7900	0.9495	0.9581	0.9934	0.9551	0.9887	0.9591	0.9644	**0.9987**
	$AUC_{(D,\tau)}$	0.2606	0.0384	0.2657	0.0294	0.3298	0.1692	0.0863	0.0597	0.4702	**0.5665**	0.1215	0.2152
	$AUC_{(F,\tau)}$	0.0660	0.0093	0.0755	0.0209	0.1214	0.0972	0.0151	0.0124	0.1320	0.2240	0.0537	**0.0006**
	AUC_{TD}	1.2545	1.0091	1.0997	0.8194	1.2793	1.1272	1.0797	1.0148	1.4588	**1.5256**	1.0858	1.2139
	AUC_{BS}	0.9279	0.9615	0.7585	0.7691	0.8281	0.8608	0.9783	0.9427	0.8567	0.7351	0.9106	**0.9981**
	AUC_{SNPR}	3.9484	4.1400	3.5171	1.4092	2.7156	1.7397	5.7093	4.8276	3.5621	2.5288	2.2615	**357.2519**
	AUC_{TDBS}	0.1946	0.0291	0.1901	0.0085	0.2083	0.0719	0.0712	0.0473	0.3382	**0.3425**	0.0678	0.2146
	AUC_{ODP}	1.1885	0.9999	1.0242	0.7985	1.1579	1.0300	1.0646	1.0024	**1.3268**	1.3016	1.0321	1.2133
Pavia	$AUC_{(D,F)}$	0.8655	0.9980	0.9964	0.9982	0.9953	0.9860	0.9973	0.99939	0.9969	0.99937	0.9973	**0.9995**
	$AUC_{(D,\tau)}$	0.1255	0.2402	0.3228	0.2071	0.3064	0.3525	0.2459	0.3284	**0.4481**	0.3961	0.3097	0.3536
	$AUC_{(F,\tau)}$	0.0684	0.0254	0.0845	0.0234	0.0211	0.0414	0.0155	0.0056	0.0971	0.0069	0.0183	**0.0005**
	AUC_{TD}	0.9909	1.2382	1.3192	1.2053	1.3017	1.3385	1.2431	1.3278	**1.4450**	1.3955	1.3070	1.3531
	AUC_{BS}	0.7971	0.9726	0.9119	0.9748	0.9742	0.9446	0.9818	0.9938	0.8998	0.9925	0.9790	**0.9990**
	AUC_{SNPR}	1.8352	9.4497	3.8208	8.8373	14.4903	8.5167	15.8649	58.8178	4.6132	57.6500	16.9087	**752.2952**
	AUC_{TDBS}	0.0571	0.2148	0.2383	0.1836	0.2852	0.3111	0.2304	0.3228	0.3510	**0.3892**	0.2914	0.3531
	AUC_{ODP}	0.9226	1.2128	1.2347	1.1818	1.2805	1.2971	1.2276	1.3222	1.3479	**1.3886**	1.2887	1.3526
Hyperion	$AUC_{(D,F)}$	0.9869	0.9843	0.9929	0.9835	0.9626	0.7938	0.9908	0.9937	0.9632	0.9432	0.9943	**0.9952**
	$AUC_{(D,\tau)}$	0.3145	0.1976	0.3693	0.2319	0.3463	0.3725	0.2091	0.2809	**0.4064**	0.3990	0.3301	0.3112
	$AUC_{(F,\tau)}$	0.0636	0.0234	0.0751	0.0437	0.0751	0.2385	0.0285	**0.0122**	0.0868	0.0893	0.0763	0.0211
	AUC_{TD}	1.3014	1.1819	1.3622	1.2154	1.3089	1.1663	1.1998	1.2745	**1.3696**	1.3421	1.3244	1.3064
	AUC_{BS}	0.9233	0.9609	0.9177	0.9399	0.8875	0.5553	0.9623	**0.9815**	0.8764	0.8539	0.9180	0.9742
	AUC_{SNPR}	4.9450	8.4398	4.9145	5.3114	4.6139	1.5616	7.3367	**23.0634**	4.6807	4.4681	4.3262	14.7775
	AUC_{TDBS}	0.2509	0.1742	0.2942	0.1882	0.2712	0.1340	0.1806	0.2687	**0.3196**	0.3097	0.2538	0.2901
	AUC_{ODP}	1.2378	1.1585	**1.2870**	1.1717	1.2338	0.9277	1.1713	1.2624	1.2828	1.2528	1.2481	1.2853
Beach-3	$AUC_{(D,F)}$	0.9781	0.9983	0.9891	0.9995	0.9927	**0.9998**	0.9991	0.9772	0.9915	0.9696	0.9930	0.9982
	$AUC_{(D,\tau)}$	0.5321	0.5108	0.4179	0.55771	0.5207	**0.6155**	0.5085	0.3533	0.5427	0.55768	0.5358	0.4305
	$AUC_{(F,\tau)}$	0.0680	0.0239	0.0559	0.0172	0.0698	0.0626	0.0092	0.0147	0.0472	0.0950	0.0696	**0.0006**
	AUC_{TD}	1.5102	1.5091	1.4070	1.5572	1.5134	**1.6152**	1.5076	1.3306	1.5342	1.5273	1.5288	1.4287
	AUC_{BS}	0.9102	0.9745	0.9332	0.9824	0.9229	0.9372	0.9900	0.9625	0.9443	0.8747	0.9234	**0.9976**
	AUC_{SNPR}	7.8287	21.3866	7.4708	32.4771	7.4583	9.8393	55.3837	24.0104	11.4985	5.8717	7.7000	**749.3484**
	AUC_{TDBS}	0.4641	0.4869	0.3620	0.5405	0.4509	**0.5529**	0.4993	0.3386	0.4955	0.4627	0.4662	0.4300
	AUC_{ODP}	1.4423	1.4853	1.3511	1.5401	1.4436	**1.5527**	1.4984	1.3159	1.4870	1.4323	1.4592	1.4281
Urban-3	$AUC_{(D,F)}$	0.7761	0.9298	0.9614	0.9577	0.9466	0.9703	0.9710	0.9755	0.9608	0.9545	0.9701	**0.9836**
	$AUC_{(D,\tau)}$	0.0978	0.1328	0.2665	0.1081	**0.4757**	0.2958	0.2379	0.1353	0.3780	0.2323	0.3686	0.3963
	$AUC_{(F,\tau)}$	0.0633	0.0464	0.0313	0.0328	0.1212	0.0785	0.0289	**0.0142**	0.0645	0.0503	0.0712	0.0147
	AUC_{TD}	0.8739	1.0626	1.2278	1.0658	**1.4223**	1.2661	1.2089	1.1108	1.3387	1.1869	1.3387	1.3799
	AUC_{BS}	0.7128	0.8835	0.9301	0.9250	0.8254	0.8918	0.9421	0.9613	0.8962	0.9042	0.8990	**0.9689**
	AUC_{SNPR}	1.5439	2.8644	8.5233	3.2980	3.9239	3.7674	8.2210	9.5574	5.8560	4.6151	5.1806	**27.0146**
	AUC_{TDBS}	0.0344	0.0864	0.2352	0.0753	0.3544	0.2173	0.2090	0.1212	0.3134	0.1820	0.2975	**0.3817**
	AUC_{ODP}	0.8106	1.0163	1.1966	1.0330	1.3011	1.1876	1.1800	1.0966	1.2742	1.1365	1.2676	**1.3652**
Urban-4	$AUC_{(D,F)}$	0.8734	0.9755	0.9833	0.9871	0.9288	0.9735	0.9713	0.9919	0.9943	0.9912	0.9865	**0.9951**
	$AUC_{(D,\tau)}$	0.0326	0.1179	0.0331	0.1050	0.0358	0.0297	0.1210	0.0369	0.1257	0.1300	0.0770	**0.1303**
	$AUC_{(F,\tau)}$	**0.0005**	0.0122	0.0013	0.0115	0.0043	0.0015	0.0179	0.0008	0.0112	0.0182	0.0065	0.0118
	AUC_{TD}	0.9060	1.0935	1.0164	1.0921	0.9646	1.0032	1.0922	1.0288	1.1200	1.1212	1.0635	**1.1254**
	AUC_{BS}	0.8729	0.9633	0.9820	0.9756	0.9245	0.9720	0.9534	**0.9911**	0.9831	0.9730	0.9800	0.9833
	AUC_{SNPR}	**68.1416**	9.6299	25.4119	9.1220	8.4031	19.8524	6.7569	43.6067	11.2181	7.1315	11.8483	11.0710
	AUC_{TDBS}	0.0321	0.1057	0.0318	0.0935	0.0315	0.0282	0.1031	0.0361	0.1145	0.1118	0.0705	**0.1185**
	AUC_{ODP}	0.9055	1.0812	1.0151	1.0806	0.9603	1.0017	1.0743	1.0280	1.1088	1.1030	1.0570	**1.1136**
Urban-5	$AUC_{(D,F)}$	0.9036	0.9712	0.9495	0.9658	0.9354	0.9331	0.9556	0.9453	0.9697	0.9409	0.9699	**0.9727**
	$AUC_{(D,\tau)}$	0.1342	0.1900	0.1799	0.1510	0.2994	0.1708	0.0952	0.1257	**0.3676**	0.3329	0.3380	0.3390
	$AUC_{(F,\tau)}$	0.0495	0.0365	0.0475	0.0453	0.0900	0.0649	0.0313	**0.0229**	0.0610	0.0578	0.0707	0.0289
	AUC_{TD}	1.0379	1.1612	1.1294	1.1168	1.2347	1.1039	1.0508	1.0710	**1.3372**	1.2737	1.3079	1.3117
	AUC_{BS}	0.8541	0.9347	0.9020	0.9205	0.8454	0.8682	0.9243	0.9223	0.9086	0.8831	0.8992	**0.9438**
	AUC_{SNPR}	2.7109	5.2098	3.7866	3.3317	3.3277	2.6295	3.0404	5.4814	6.0234	5.7582	4.7802	**11.7385**
	AUC_{TDBS}	0.0847	0.1535	0.1324	0.1057	0.2094	0.1058	0.0639	0.1028	0.3065	0.2750	0.2673	**0.3102**
	AUC_{ODP}	0.9884	1.1247	1.0819	1.0715	1.1448	1.0389	1.0194	1.0480	1.2762	1.2159	1.2372	**1.2828**
Airport-4	$AUC_{(D,F)}$	0.9164	0.9845	0.9321	0.9627	0.9867	0.9827	0.9804	0.9680	0.9840	0.9788	0.9477	**0.9914**
	$AUC_{(D,\tau)}$	0.1265	0.3238	0.2477	0.0666	0.4167	0.3702	0.0519	0.2470	0.5149	**0.6145**	0.4557	0.4848
	$AUC_{(F,\tau)}$	0.0329	0.0291	0.0896	0.0165	0.0890	0.1130	0.0118	0.0303	0.1392	0.1012	0.1160	**0.0071**
	AUC_{TD}	1.0429	1.3083	1.1798	1.0293	1.4034	1.3529	1.0323	1.2150	1.4988	**1.5933**	1.4034	1.4763
	AUC_{BS}	0.8835	0.9554	0.8424	0.9462	0.8977	0.8698	0.9686	0.9377	0.8448	0.8775	0.8317	**0.9844**
	AUC_{SNPR}	3.8433	11.1146	2.7638	4.0344	4.6823	3.2771	4.4021	8.1544	3.6984	6.0702	3.9280	**68.4670**
	AUC_{TDBS}	0.0936	0.2947	0.1581	0.0501	0.3277	0.2572	0.0401	0.2167	0.3756	**0.5133**	0.3397	0.4778
	AUC_{ODP}	1.0100	1.2792	1.0902	1.0128	1.3144	1.2399	1.0205	1.1847	1.3596	**1.4921**	1.2874	1.4692

2) Tradeoff Parameters α, β, and γ: The detection performances under different tradeoff parameters α, β, and γ are displayed in Fig. 17(b)–(d), respectively. The value ranges for α and γ are $\{1e\text{-}5, 1e\text{-}4, 1e\text{-}3, 1e\text{-}2, 1e\text{-}1, 0.5, 1\}$ and $\{0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1, 2.5, 5, 10\}$, while the value range for β is from 0.01 to 0.09 with an interval of 0.01.

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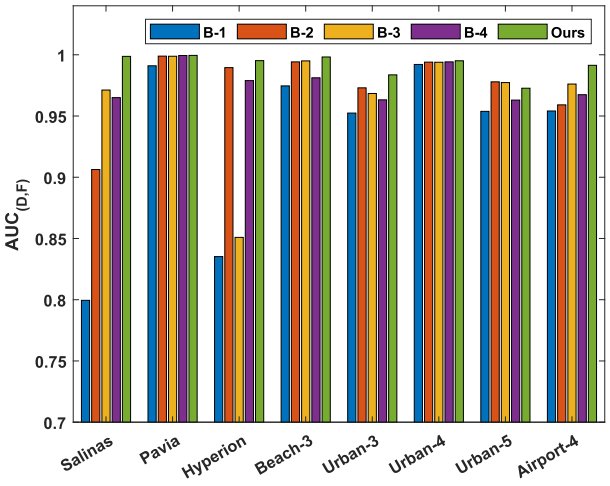


Fig. 18. $\text{AUC}_{(D,F)}$ of the MERAETV model with different low-rank approximation methods.

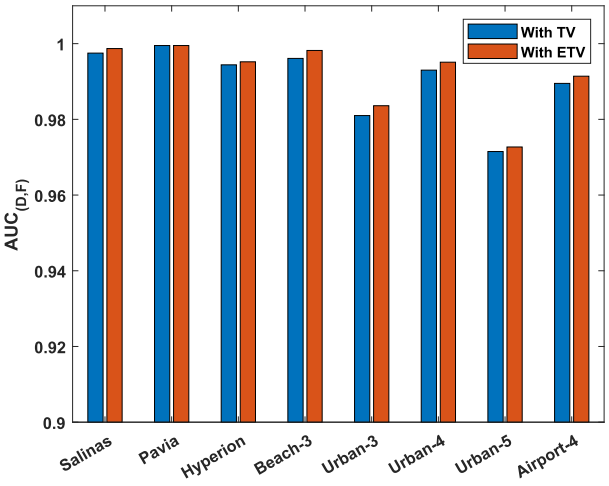


Fig. 19. $\text{AUC}_{(D,F)}$ comparison of MERAETV with TV and with ETV.

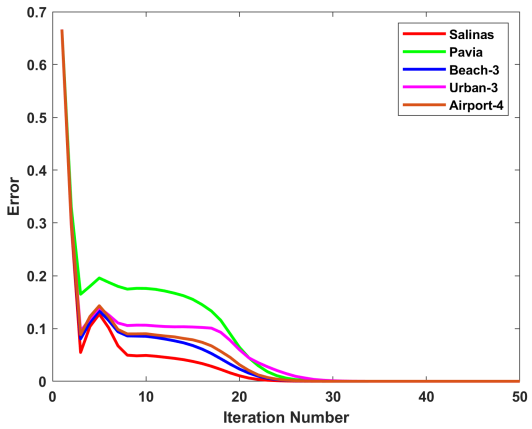


Fig. 20. Convergence curves of MERAETV algorithm on five datasets.

As can be seen from Fig. 17(b), when $\alpha = 1e\text{-}2$, the proposed MERAETV algorithm achieves satisfactory detection performance on the Pavia, Beach-3, Urban-5, and Airport-4 datasets. When α takes 1, $1e$-5, $1e$-3, and 0.5, respectively, the detection performance reaches the best on the Salinas, Hyperion, Urban-3, and Urban-4 datasets. As exhibited in Fig. 17(c), when $\beta \in [0.01, 0.06]$, the proposed MERAETV algorithm achieves a stable detection performance on Salinas, Pavia, Hyperion, Beach-3, Urban-4, and Airport-4 datasets. When β takes values of 0.05 and 0.09 on the Urban-3 and Urban-5 datasets, respectively, the optimal detection performance is achieved. It can be observed from Fig. 17(d) that when $\gamma \in [0.01, 10]$, the MERAETV algorithm achieves a stable and exciting $\text{AUC}_{(D,F)}$ on the Salinas, Beach-3, and Airport-4 datasets. Furthermore, when $\gamma \in [0.5, 10]$, a relatively stable $\text{AUC}_{(D,F)}$ is obtained on all datasets.

G. Ablation Study

This section aims to study the impact of MERA decomposition, designed ETV regularization, and decomposed noise term $\mathcal{N}$ on the proposed MERAETV model. The $\text{AUC}_{(D,F)}$ values of the experimental results are shown in Figs. 18 and 19 and Table IV, respectively. Among them, B-1, B-2, B-3, and B-4 in Fig. 18 represent the corresponding models after replacing the low-rank MERA approximation in the MERAETV model with low-rank CP approximation, low-rank Tucker approximation, low-rank t-SVD approximation, and low-rank TRD approximation, respectively. In order to facilitate the experiment, the parameter values are consistent with those set in the above experiment.

1) Effectiveness of MERA Decomposition: In order to verify the effectiveness of MERA decomposition, we, respectively, replace the low-rank MERA approximation in the MERAETV model with low-rank CP approximation, low-rank Tucker approximation, low-rank t-SVD approximation, and low-rank TRD approximation to achieve anomaly detection. The experimental results are displayed in Fig. 18. It can be easily noticed from Fig. 18 that, except for the Urban-5 dataset, the proposed MERAETV algorithm achieves optimal $\text{AUC}_{(D,F)}$ on other datasets. This shows that MERA decomposition can fully exploit the spectral–spatial correlation of the background, thereby improving detection performance.

2) Effectiveness of ETV Regularization: In order to verify the effectiveness of the designed ETV regularization term, Fig. 19 shows the $\text{AUC}_{(D,F)}$ of MERAETV with TV and with ETV. From Fig. 19, it can be found that the MERAETV model with ETV outperforms the MERAETV model with TV on almost all datasets, which fully presents that the detection performance can be improved by simultaneously exploiting the piecewise-smooth of the background in the spectral and spatial domains.

3) With and Without Noise Term: To study the impact of the decomposed noise term $\mathcal{N}$ on anomaly detection performance, Table IV reports the detection performance and running time of MERAETV with and without $\mathcal{N}$ on eight HSI datasets. As can be seen from Table IV, compared with MERAETV without $\mathcal{N}$, except for the Urban-4 dataset, the evaluation indicators $\text{AUC}_{(F,\tau)}$, AUC_{BS}, and AUC_{SNPR} and time of MERAETV on the other seven datasets are all optimal. Although MERAETV without $\mathcal{N}$ has better $\text{AUC}_{(D,F)}$ than MERAETV on Salinas, Hyperion, Beach-3, Urban-5, and Airport-4 datasets, the difference between the two results is

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TABLE IV
COMPARISON OF DETECTION PERFORMANCE OF MERAETV WITH AND WITHOUT NOISE TERM $\mathcal{N}$ ON EIGHT DATASETS

Datasets	Methods	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{BS}	AUC_{SNPR}	Time
Salinas	MERAETV without $\mathcal{N}$	**0.9993**	**0.7348**	0.0608	0.9385	12.0819	43.8914
	MERAETV	0.9987	0.2152	**0.0006**	**0.9981**	**357.2519**	**37.4932**
Pavia	MERAETV without $\mathcal{N}$	**0.9995**	**0.4735**	0.0162	0.9833	29.2062	14.6341
	MERAETV	**0.9995**	0.3536	**0.0005**	**0.9990**	**752.2952**	**13.1105**
Hyperion	MERAETV without $\mathcal{N}$	**0.9963**	**0.3761**	0.0469	0.9495	8.0268	37.5754
	MERAETV	0.9952	0.3112	**0.0211**	**0.9742**	**14.7775**	**32.8677**
Beach-3	MERAETV without $\mathcal{N}$	**0.9983**	**0.5583**	0.0346	0.9637	16.1543	87.2698
	MERAETV	0.9982	0.4305	**0.0006**	**0.9976**	**749.3484**	**63.5621**
Urban-3	MERAETV without $\mathcal{N}$	0.9828	**0.4703**	0.0518	0.9310	9.0870	46.8016
	MERAETV	**0.9836**	0.3963	**0.0147**	**0.9689**	**27.0146**	**40.7522**
Urban-4	MERAETV without $\mathcal{N}$	**0.9951**	**0.1303**	**0.011766**	**0.9833**	**11.0744**	42.0585
	MERAETV	**0.9951**	**0.1303**	0.011769	**0.9833**	11.0710	**41.9134**
Urban-5	MERAETV without $\mathcal{N}$	**0.9758**	**0.3629**	0.0504	0.9254	7.2068	74.4573
	MERAETV	0.9727	0.3390	**0.0289**	**0.9438**	**11.7385**	**53.6309**
Airport-4	MERAETV without $\mathcal{N}$	**0.9920**	**0.6530**	0.0942	0.8978	6.9303	21.0984
	MERAETV	0.9914	0.4848	**0.0071**	**0.9844**	**68.4670**	**19.4164**

TABLE V
RUNNING TIME (IN SECONDS) OF VARIOUS HAD ALGORITHMS ON EIGHT DATASETS

Datasets	LRX	FrFE	CRD	RPCA	GTVLRR	LSDM-MoG	FEBPAD	GAED	PCA-TLRSR	PTA	LARTVAD	Ours
Salinas	8.7301	37.9223	3.2701	4.8333	135.4833	11.3394	**3.1766**	81.5689	4.1727	26.4825	30.6687	37.4932
Pavia	1.9015	8.7888	1.4339	**0.48887**	102.7125	3.2134	0.9833	21.332	8.1025	7.797	25.3915	13.1105
Hyperion	3.8872	15.5184	1.7296	**0.56717**	83.716	2.9107	1.1314	19.6767	1.1713	9.5339	26.2062	32.8677
Beach-3	2.8678	18.3201	**0.63314**	1.3549	92.7809	5.9302	1.5067	22.3223	5.3481	12.9398	18.6503	63.5621
Urban-3	4.4971	23.1365	3.5527	1.4816	98.2485	3.2001	1.4835	22.7934	**1.2214**	14.6027	27.2722	40.7522
Urban-4	3.0811	20.5541	2.048	3.1001	54.3974	9.5161	**1.6588**	35.1331	2.152	12.946	18.6171	41.9134
Urban-5	3.5637	24.9601	3.5844	3.1346	88.6196	3.6891	1.693	23.6167	**1.1641**	13.702	27.0009	53.6309
Airport-4	1.9528	24.68	2.3799	**1.4651**	96.8432	3.4836	1.5457	22.6874	5.8174	14.1342	30.0859	19.4164
Average	3.8102	21.735	2.339	2.0532	94.1002	5.4103	**1.6474**	31.1413	3.6437	14.0173	25.4866	37.8433

small. Overall, the presence of the noise term $\mathcal{N}$ is beneficial for extracting anomalies from the background.

H. Convergence and Running Time

1) Convergence Analysis: The error curves of the MERAETV model on five test datasets are displayed in Fig. 20. The error, as defined by (30), exhibits a gradual decrease in error curves with an increasing number of iterations until it ultimately converges to zero.

2) Running Time: The computational complexity of various algorithms is assessed by recording the running time of each competing algorithm in Table V. It can be found that the average running time of the FEBPAD algorithm is the shortest. On the contrary, the GTVLRR algorithm obtains the highest average running time, followed by the proposed MERAETV algorithm. Overall, our proposed MERAETV model achieves the optimal detection performance in a relatively large average running time.

VI. CONCLUSION

This article integrates the MERA decomposition, the ETV regularization, the group sparsity constraint, and the Gaussian noise term to design a novel MERAETV model for HAD. Specifically, MERA decomposition and ETV regularization are used to fully exploit the spectral–spatial correlation and piecewise-smooth of the background tensor. In addition, the group sparsity constraint and Gaussian noise term are added to the MERAETV model to better extract anomalies from the background. Finally, the ADMM is used to optimize and solve the MERAETV model. Comprehensive performance comparisons on eight datasets verify that our proposed approach outperforms existing state-of-the-art approaches.

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